



Inspiring Excellence

CSE 251

Electronic Devices and Circuits

Lecture 7

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Outline

- **Linear** IV characteristics.
- **Non-linear** IV characteristics
- **Piecewise Linear Approximation of Non-linear**
- Semiconductor Devices: pn junction Diode
- **pn** junction Diode IV Characteristics: Circuit Model
 - Ideal Diode Model:
 - Constant Voltage Drop (CVD) Model:
 - Constant Voltage Drop with Resistor (CVD+R) Model
 - Exponential Model (Shockley Model):

Current-Voltage (I-V) Characteristics

- I-V characteristic defines the relationship between the **current flow (through), I** and **voltage (across), V** an electronic device or element.
- A tool for understanding the operation of the circuit element.
- The Current-Voltage (I-V) characteristics are found by evaluating the **response** of a device/element under different **excitation** conditions. The behavior of a device depends on the **applied excitation** and can change if the excitation changes.

Type of (I-V) Characteristics

1. Linear Devices/Elements: The Current-Voltage relationship is linear i.e. the current through the element is a **linear function** of the applied voltage across it. The relationship can be characterized by:

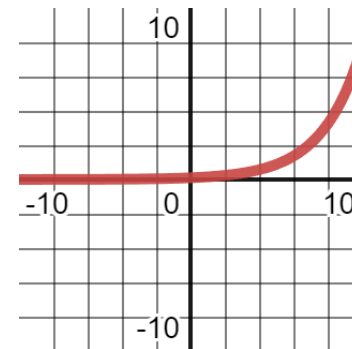
$$I = kV$$

2. Non-Linear Devices/Elements: The Current-Voltage relationship is Non-linear i.e., the current through the element is a **nonlinear function** of the applied voltage across it.

$$I = k\sqrt{V}$$

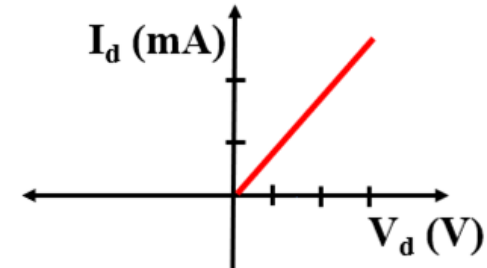
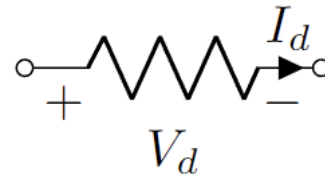
$$I = kV^2$$

$$I = kV^3$$

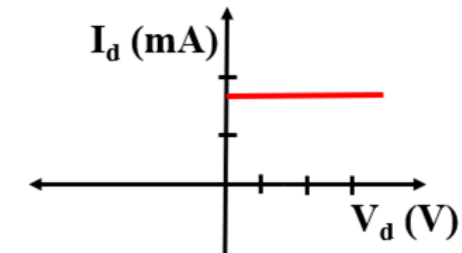
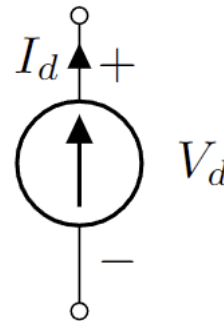


Linear IV Char.

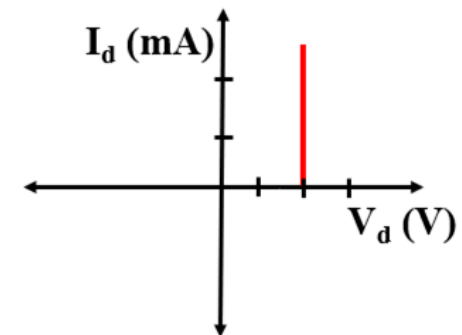
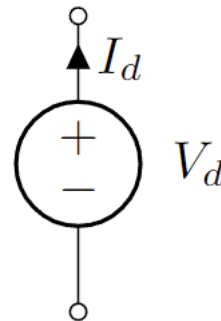
- Resistors



- Current Source

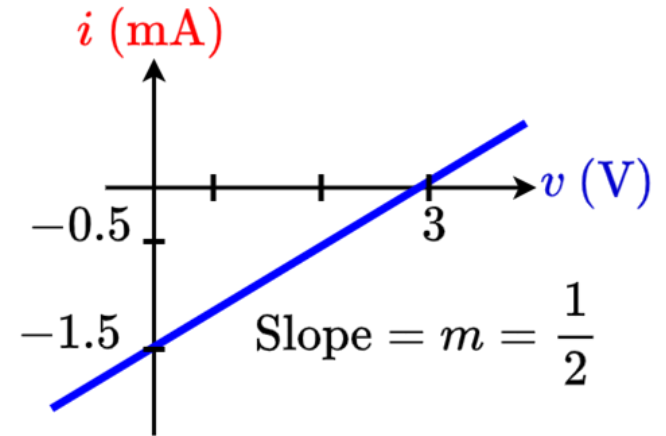
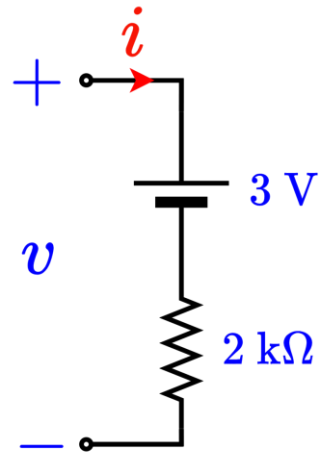


- Voltage Source

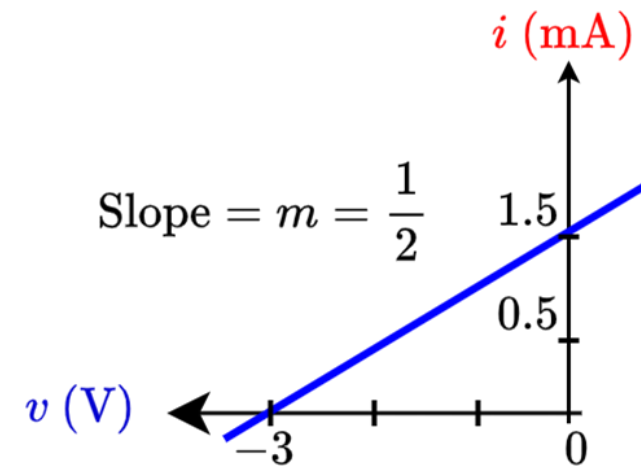
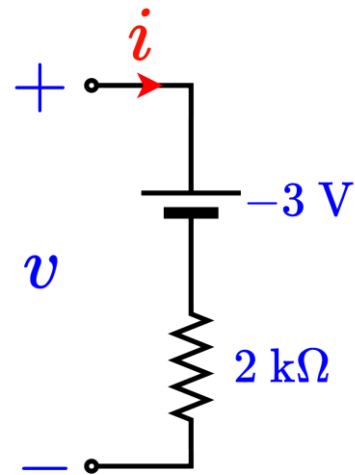


Linear IV Char.

Voltage source in series with a resistor

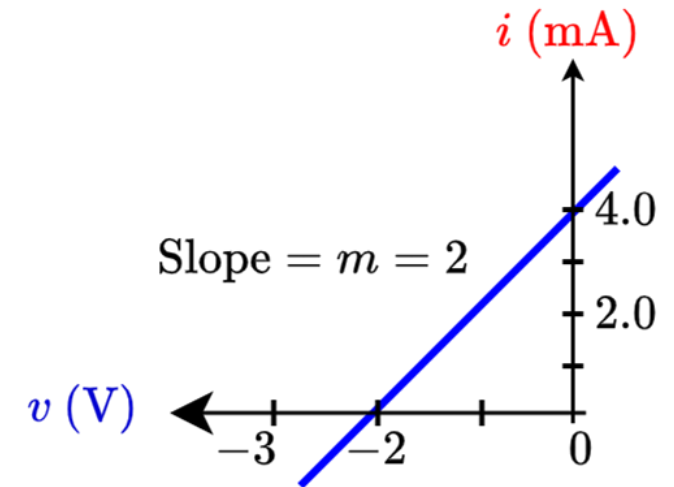
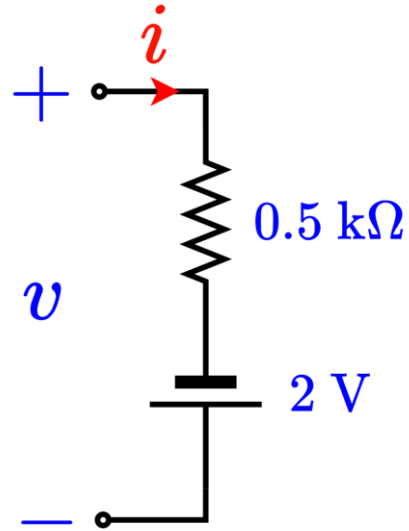


Voltage source in series with a resistor

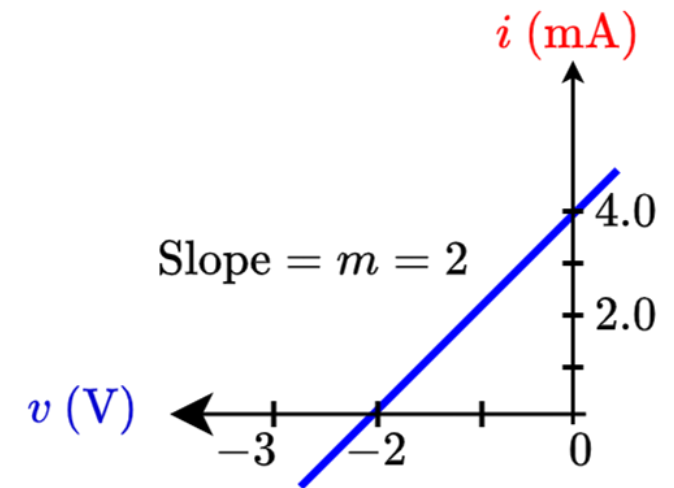
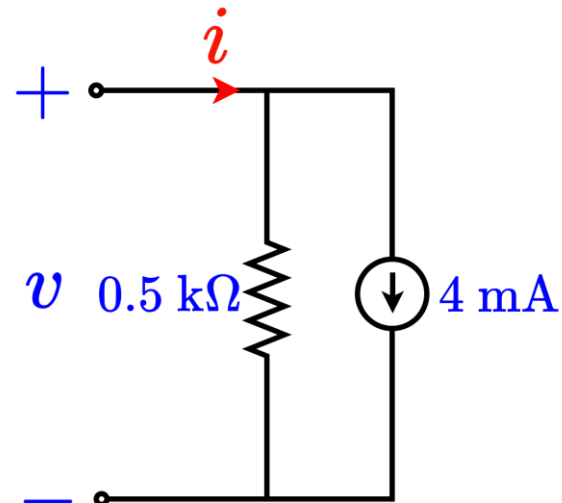


Linear IV Char.

Voltage source in series with a resistor



Current source parallel to a resistor

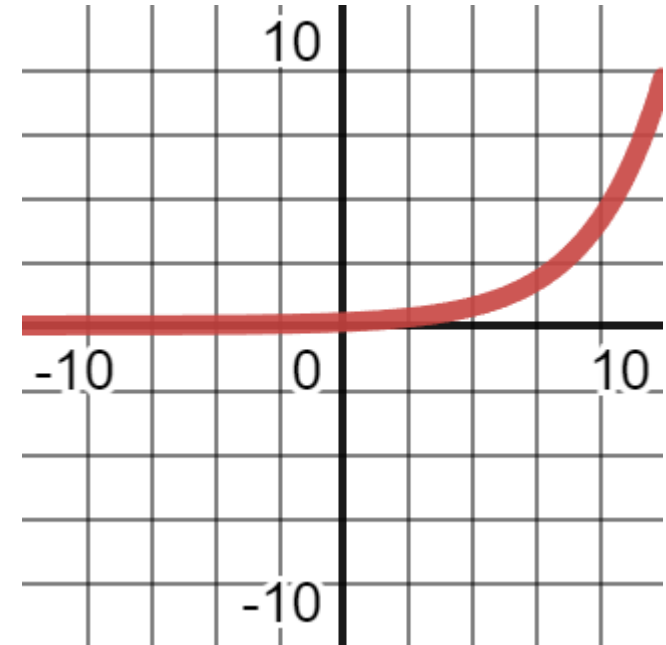


Non-Linear I-V Char.

-

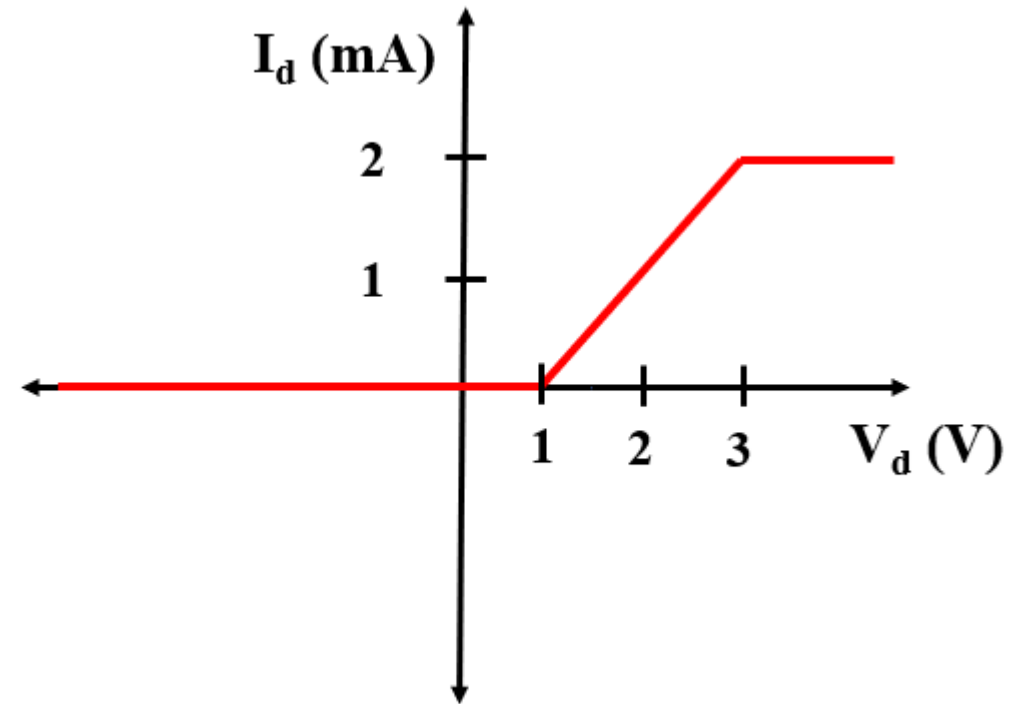
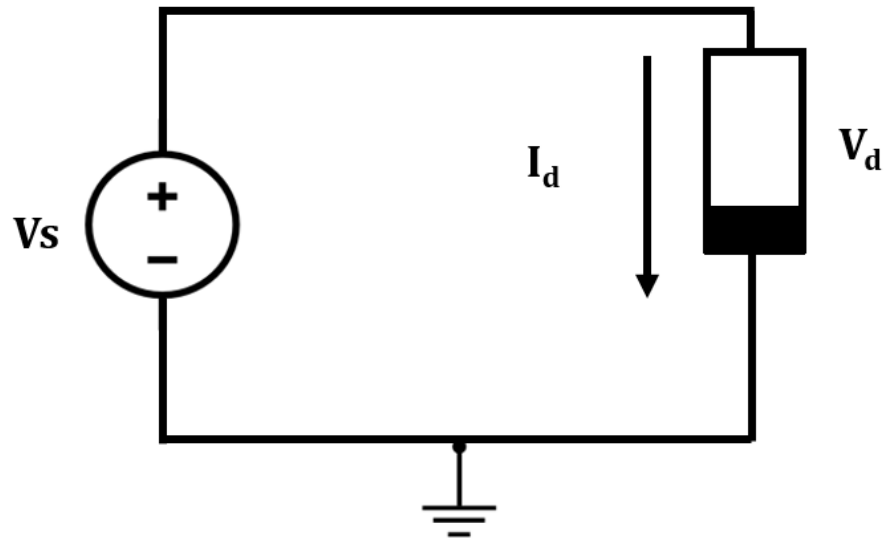
$$I = kV \quad I = kV^2 \quad I = A \cdot \exp\left(\frac{V}{b}\right)$$

$$y = mx \quad y = ax^2 \quad y = A \cdot \exp\left(\frac{x}{b}\right)$$



Non-Linear I-V Char.

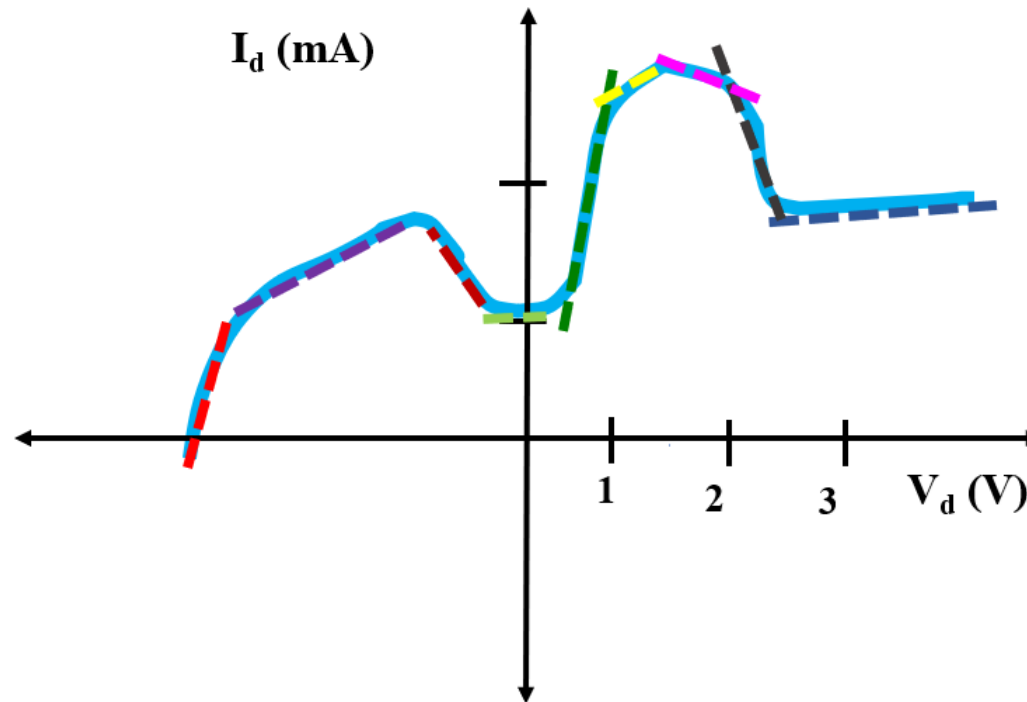
Example:



Piecewise linear function is a **Non-linear** function.

Non-Linear I-V Char.: Piecewise Linear Model

- Simplifying non-linear IV characteristics by piecewise linear parts.
- Non-linear functions are usually approximated by a series of linear segments that follow the tangent of the non-linear segment as can be seen from the following figure.



Semiconductor Materials

- Semiconductors are special class of materials (as opposed to conductors and insulators) that fall **between conductors and insulators** in terms of their electrical conductivity.
- By making simple material changes, such as **doping**, these materials can achieve remarkably precise control over electron flow.
- This ability to **control electron flow** makes semiconductors the optimal candidate for building non-linear electrical devices, where electron (current) flow is not always just proportional to the applied voltage.
- As these non-linear devices offer exceptional control over electron flow, the class of devices and circuits are branched under the umbrella term **“Electronics”**

Semiconductor Materials

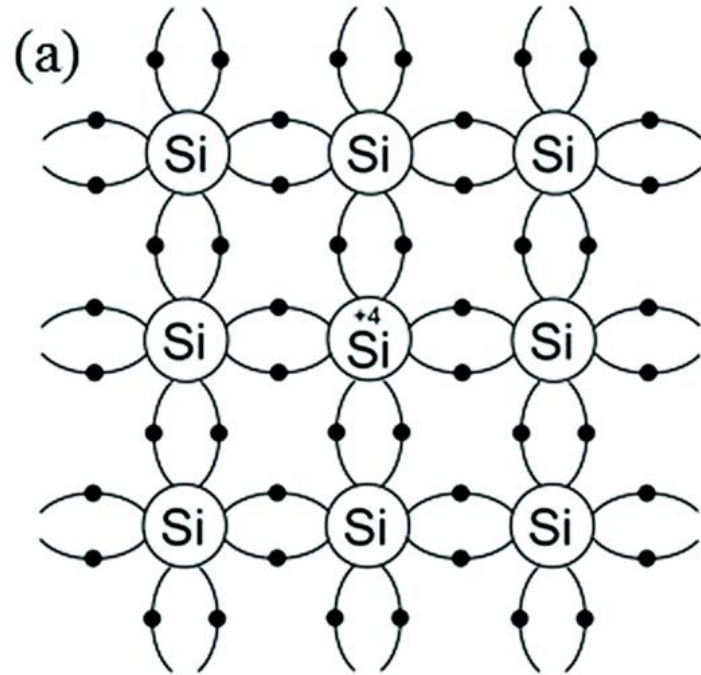
What is doping?

Doping is to deliberately inject atomic level “impurities” into a material (such as an “**intrinsic**” or **pure semiconductor** such as **Silicon**). These atoms are called **dopants**.

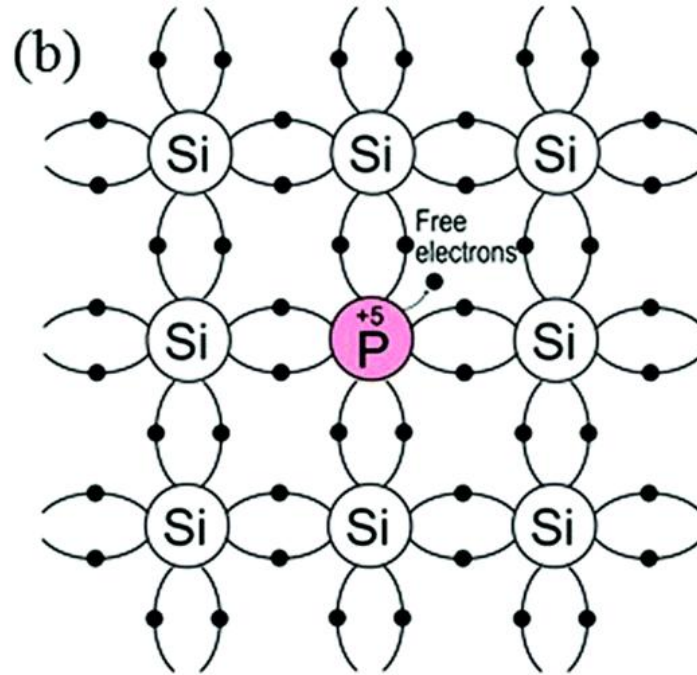
Usually, the concentration of the “**dopant**” atoms is much less than the actual concentration of the host material.

Since the atoms being injected are different from the actual material atom, they are considered impurities. A **doped** semiconductor is also known as “**Extrinsic**” semiconductor.

Semiconductor Materials

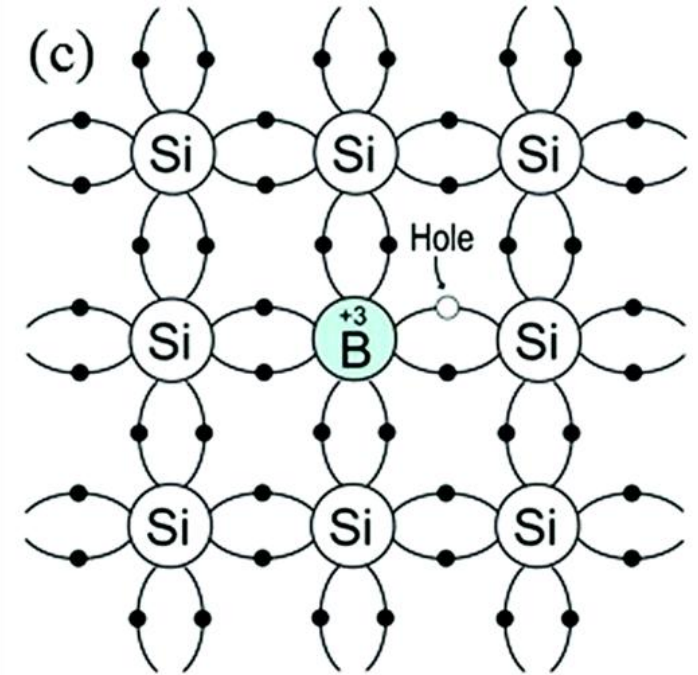


Intrinsic



n-type

n –type doped → Si lattice is riddled with **pentavalent atoms**

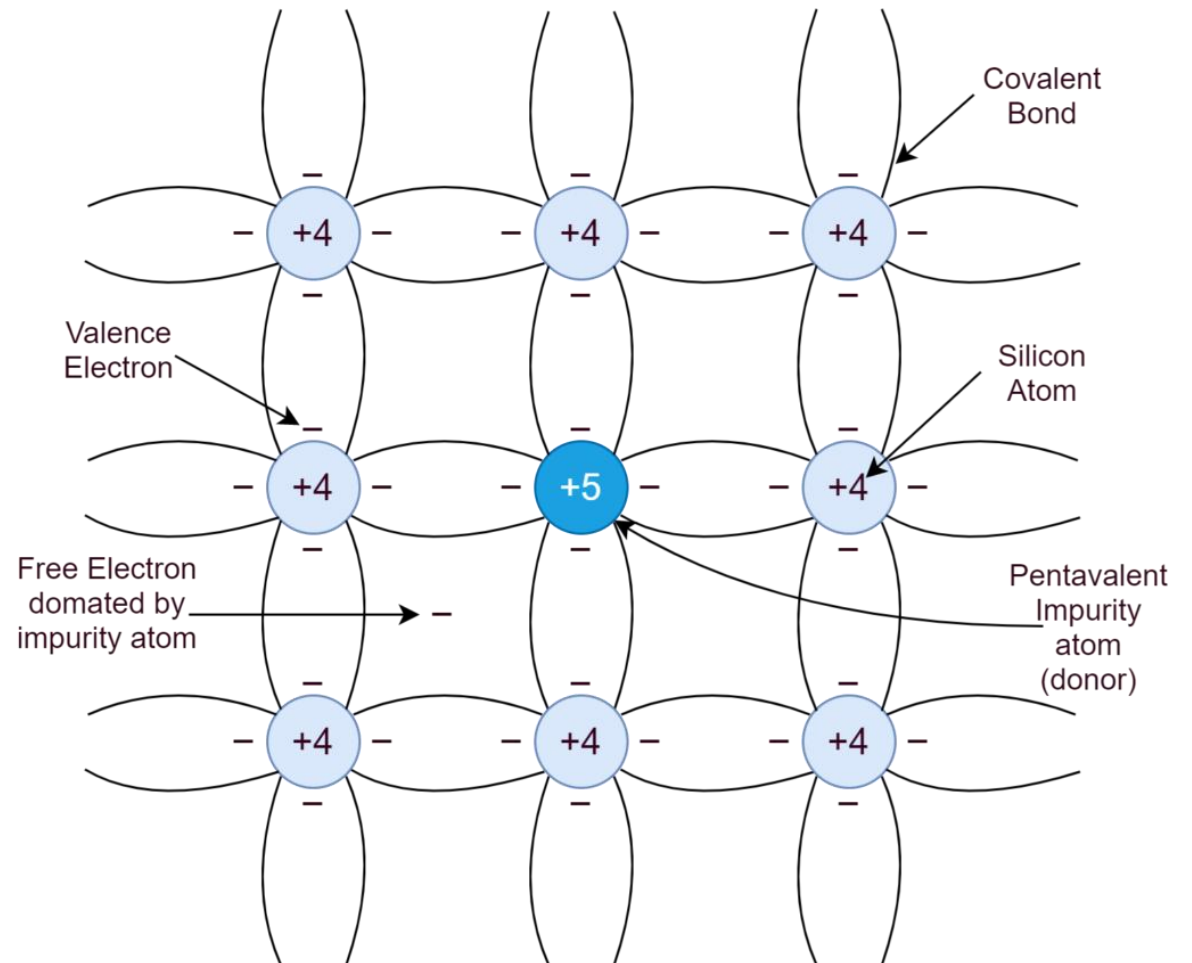


p-type

p –type doped → Si lattice is riddled with **trivalent atoms**

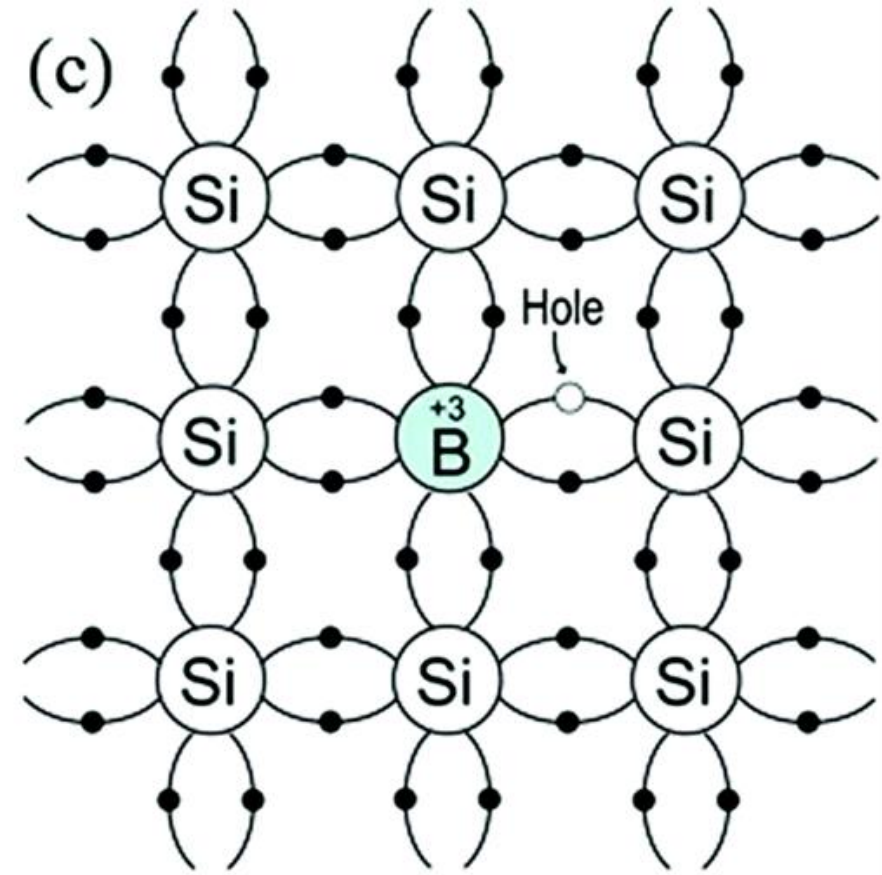
n-type doping

- **n-type materials** are doped with pentavalent atoms.
- **Pentavalent** atoms have one more electron than the surrounding **tetravalent Si** atoms.
- This lone extra electron acts as a **mobile electron** that can flow easily as it is not bonded to any atom.



p-type doping

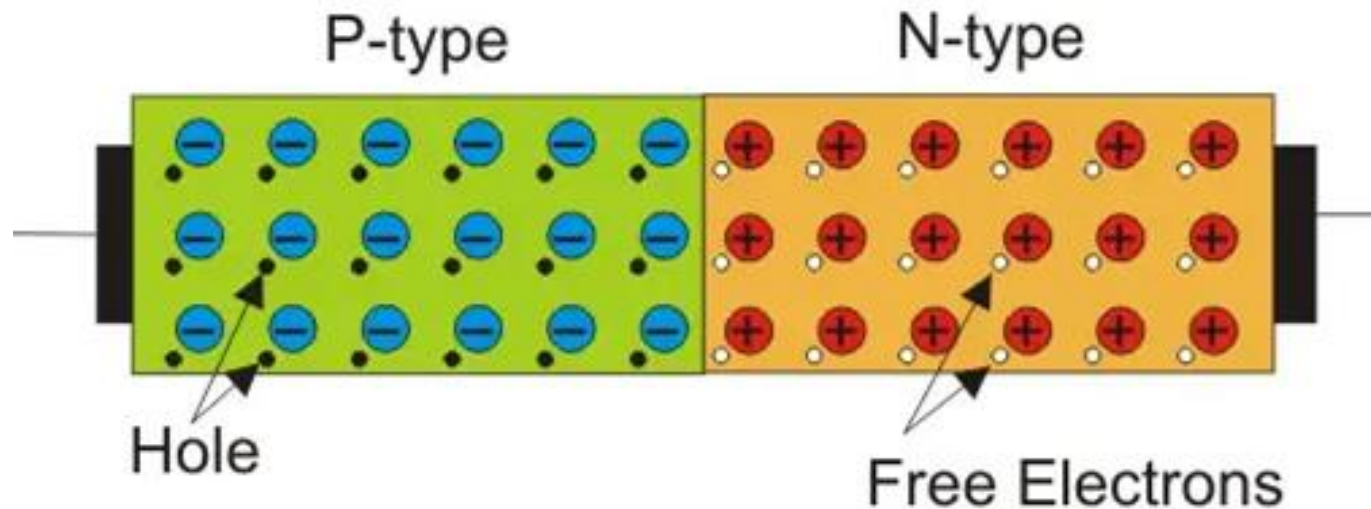
- p-type materials are doped with trivalent atoms.
- Trivalent atoms have one less electron than the surrounding tetravalent Si atoms.
- This absence of an electron (termed as a hole) acts as can flow easily from atom to atom across the material as it is not bonded to any atom. The flow of holes is opposite to the actual flow of electrons.



p-type

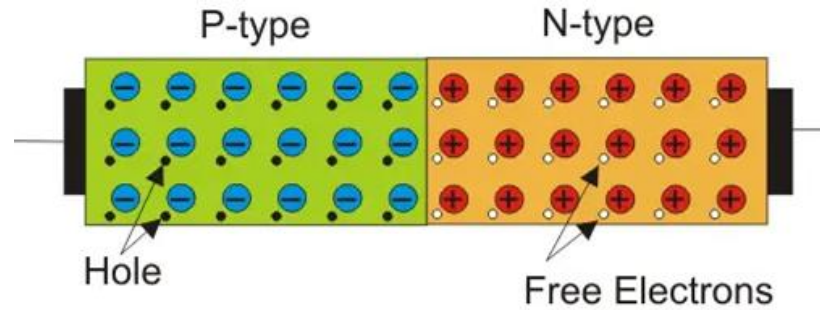
Semiconductor Devices: Diode

- **Diode** is the most basic semiconductor device.
- It is made by doping an intrinsic semiconductor (Si) half as **p-type** and the other half as **n-type**.

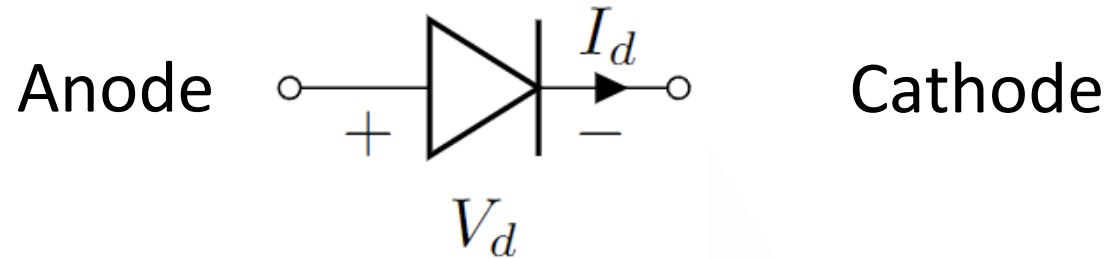


The **pn** junction diode: Physical characteristics

Internal Structure



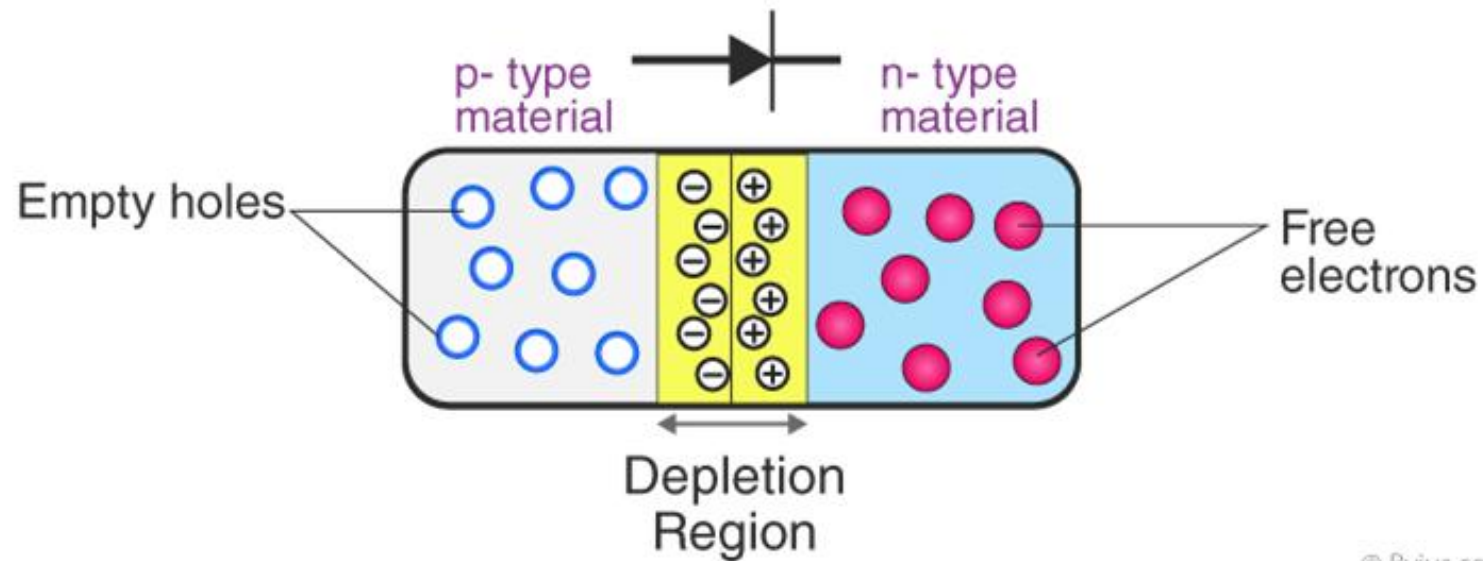
Circuit Schematic



Real device image



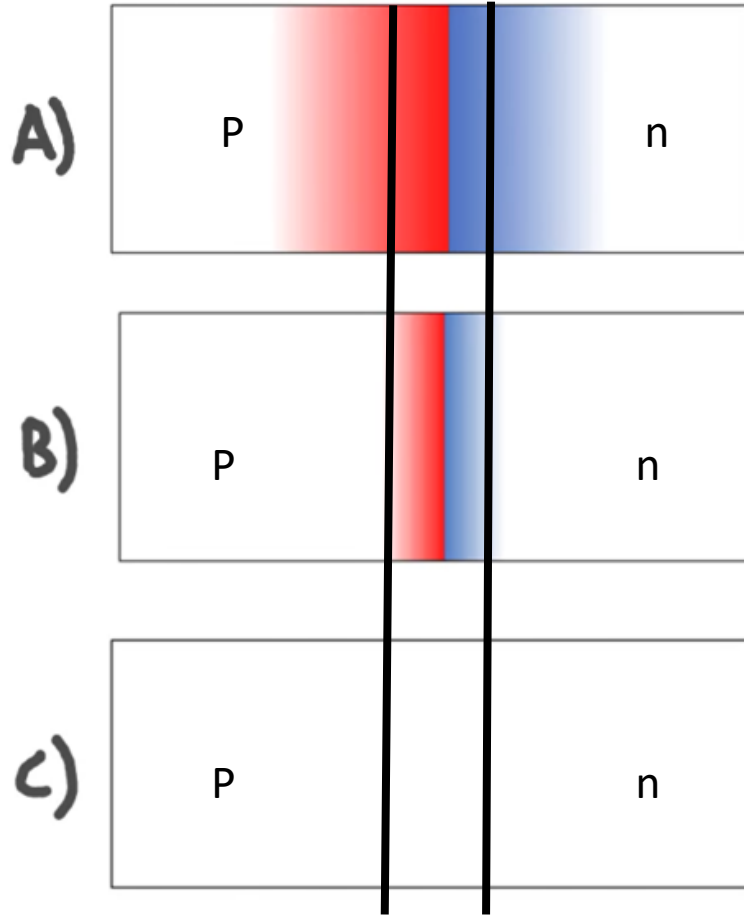
p-n junction



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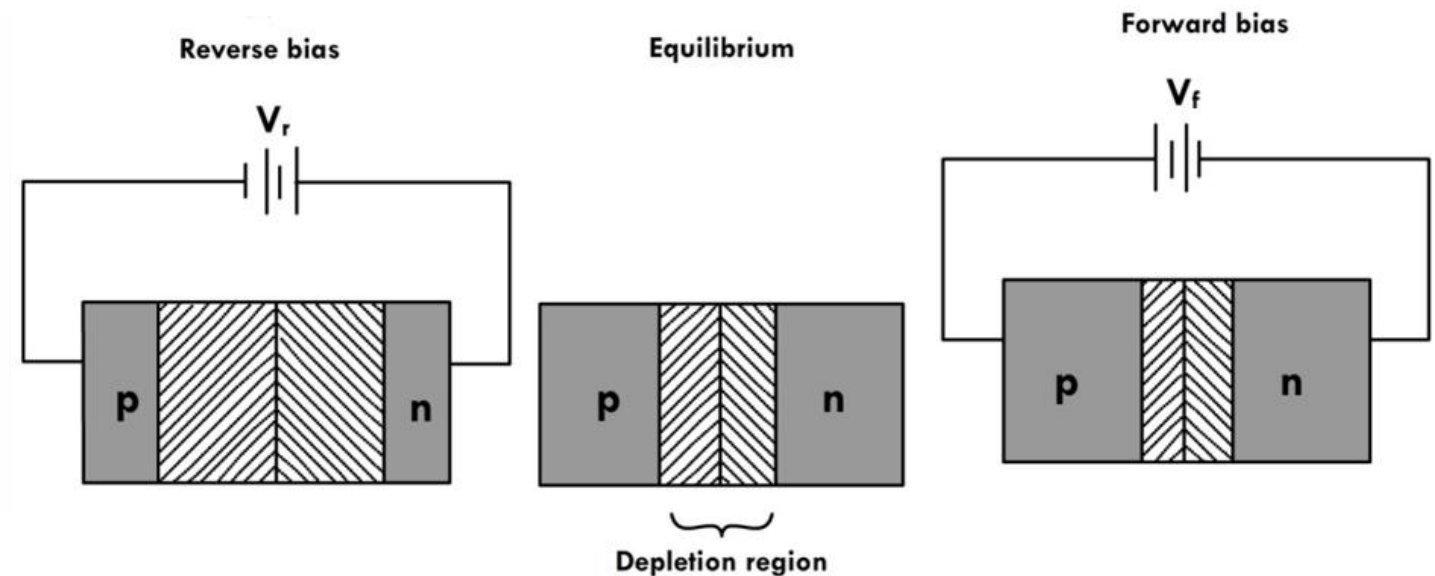
Threshold voltage- for silicon about 0.7V
- for germanium about 0.3V.

The pn junction diode: Modes of operation



Depletion Region

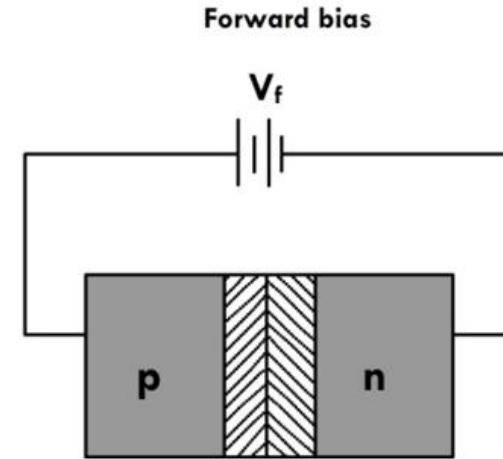
- A) Reverse Bias
- B) **No Bias**
- C) Forward Bias



The **pn** junction diode: 2 Modes of operation

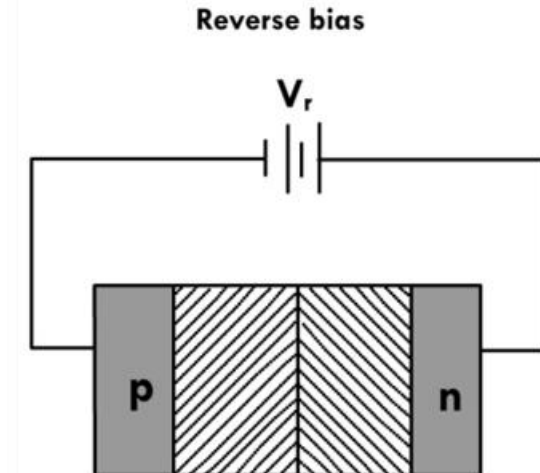
1. Forward Bias (FB):

- Depletion Region is **constricted**.
- **Allow** electrons to flow through the junction
- Ideally acts as a **short circuit**



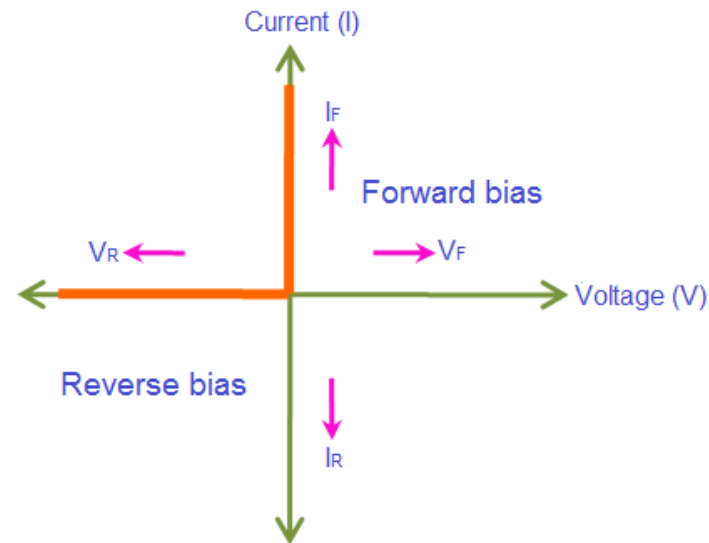
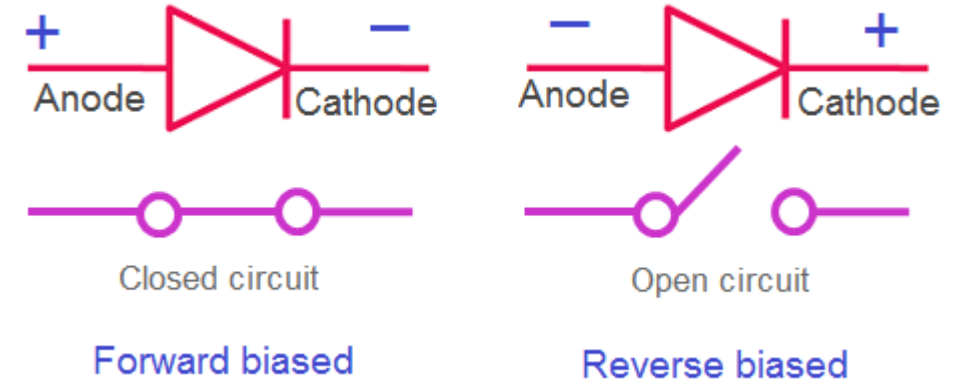
1. Reverse Bias (RB):

- Depletion Region is **expanded**.
- Does not allow electron flow through the junction.
- Ideally acts as an **open circuit**

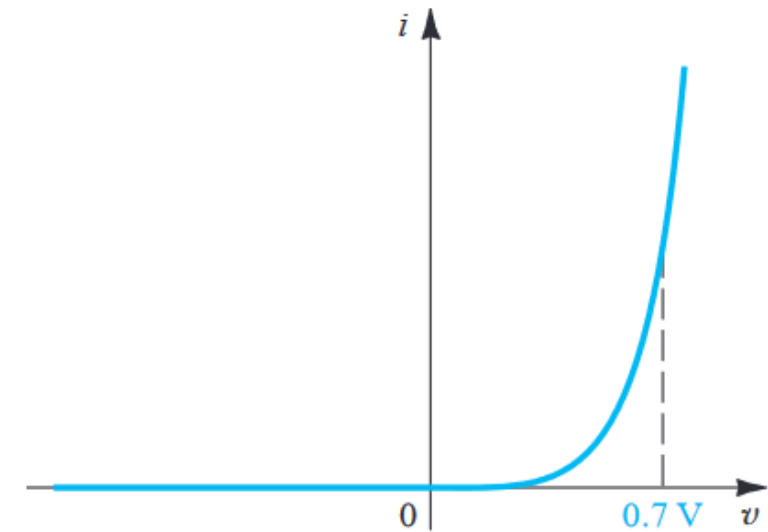


Diode Circuit Models

Ideal Versus Real Diode



Constant Voltage Drop
(CVD) Model



Ideal Model

Real / Shockley Model

Low Accuracy
Simple

CVD with resistance
(CVD+R) Model

Increased Accuracy
More Complex

Diode Circuit Models

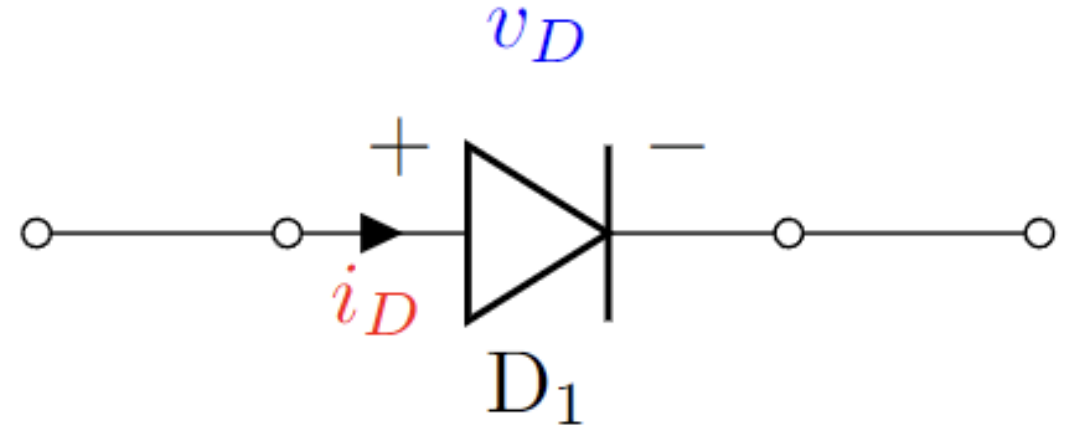
1. **Ideal Diode** Model:
2. **Constant Voltage Drop (CVD)** Model:
3. **Constant Voltage Drop with Resistor (CVD+R)** Model
4. **Exponential** Model (Shockley Model):

Piecewise
Linear IV
Models

v_D : Total Voltage Across diode
 v_d : AC component of the Voltage
 V_d : DC component of Voltage

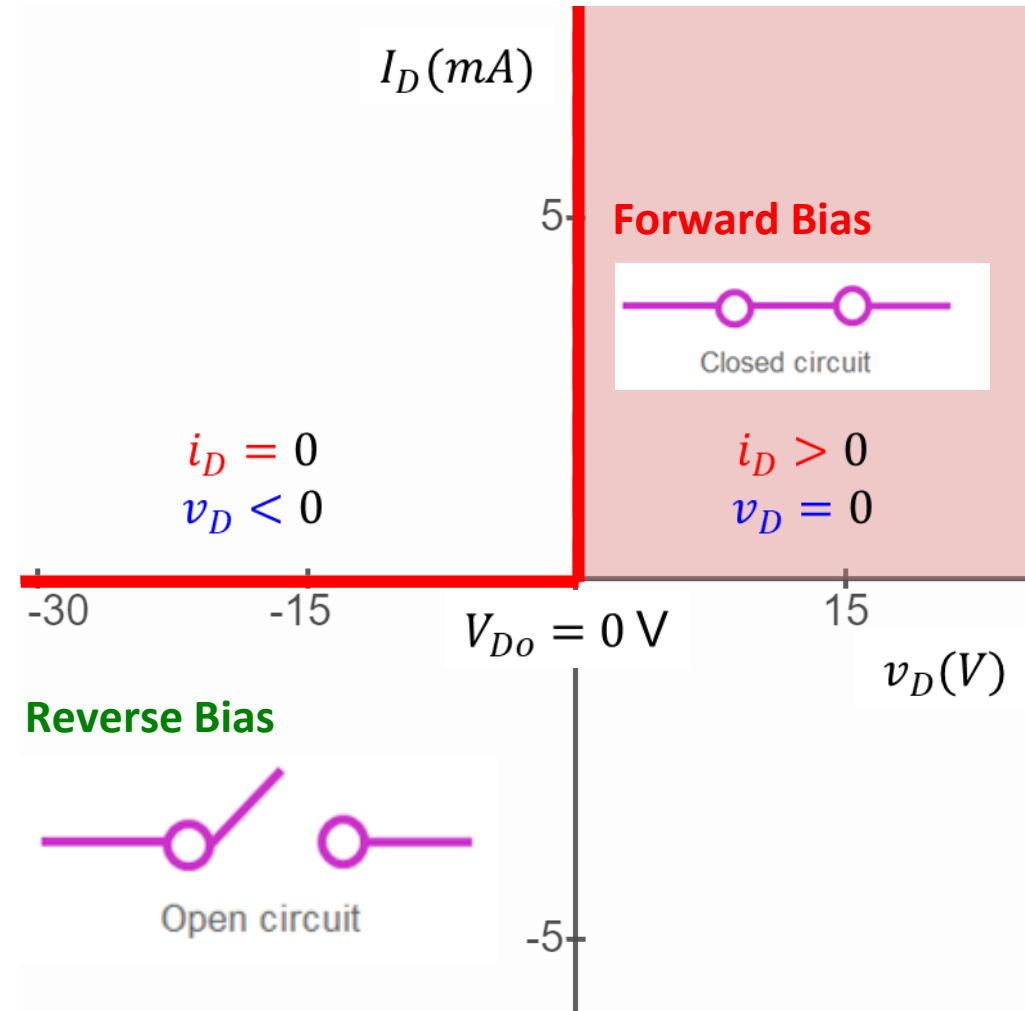
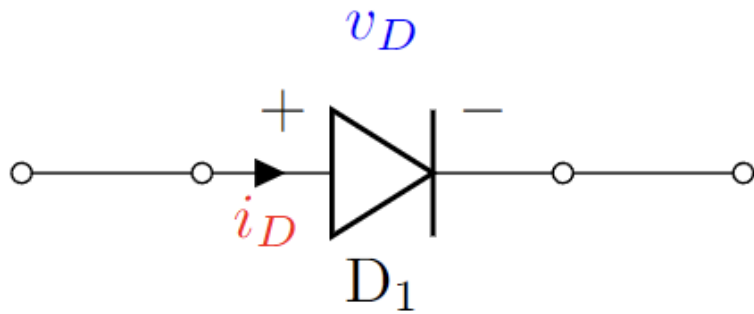
V_{D0} : Diode Cut-off voltage

i_D : Total current through diode (Anode to Cathode)



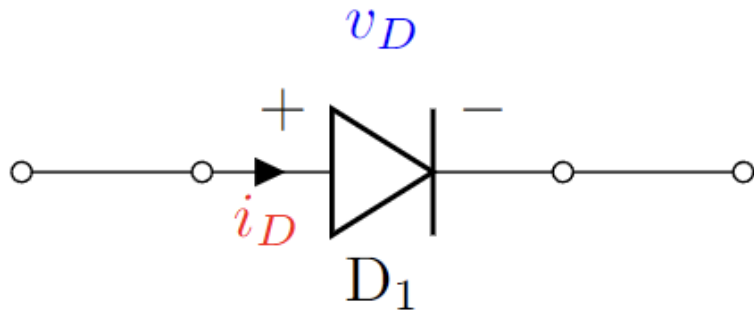
Diode Models

1. Ideal Diode Model:



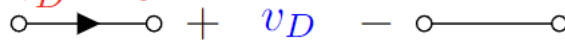
Diode Models

2. Constant Voltage Drop (CVD) Model:

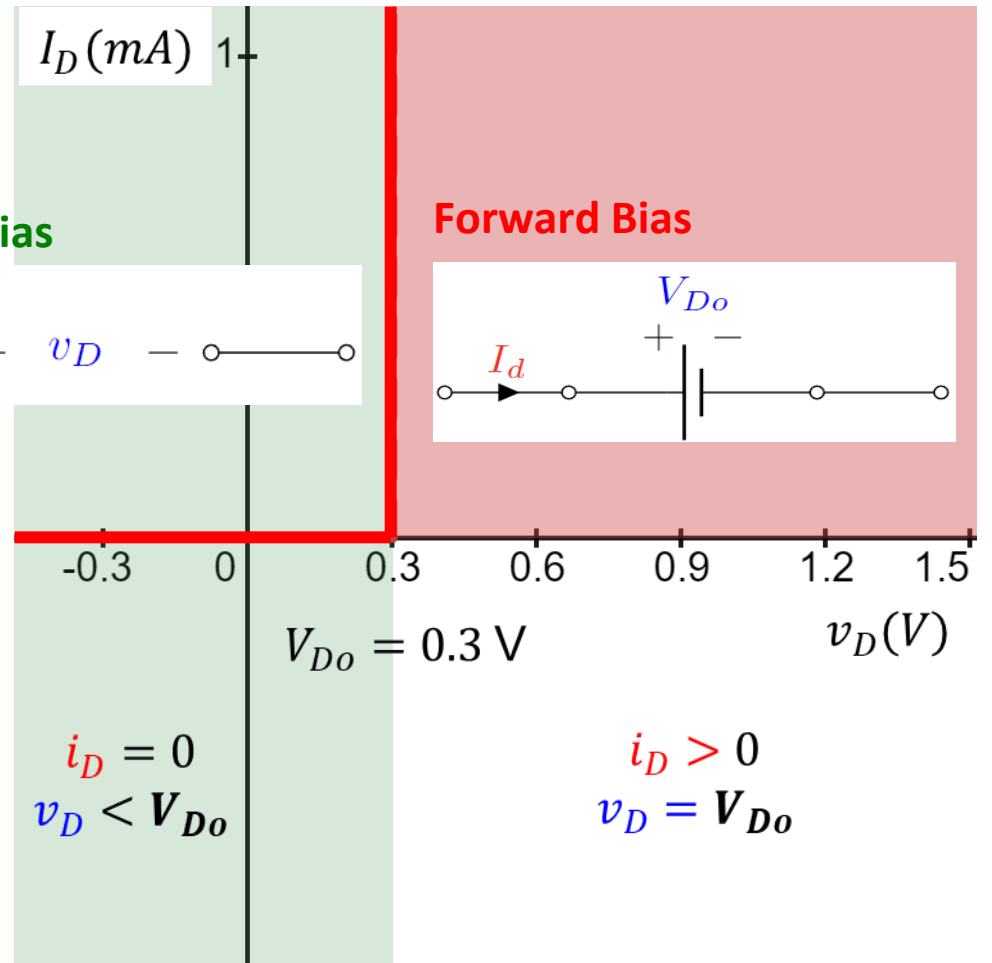
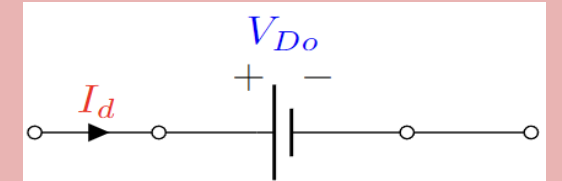


Reverse Bias

$$i_D \approx 0$$



Forward Bias

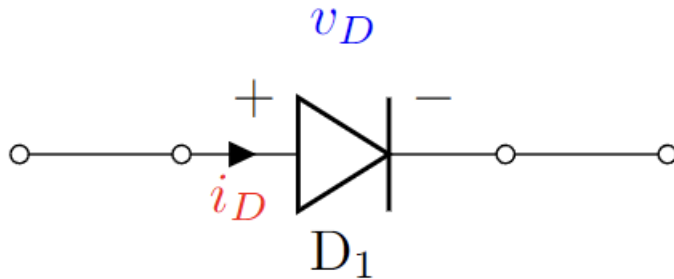


v_D : Total Voltage Across diode

V_{Do} : Diode Cut-off voltage (0.3 V here)

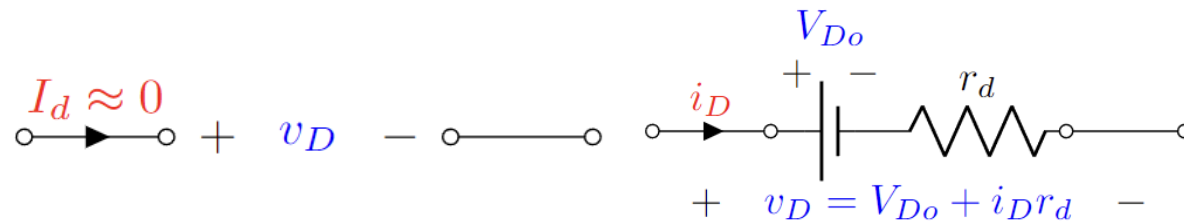
Diode Models

3. CVD+R Model:



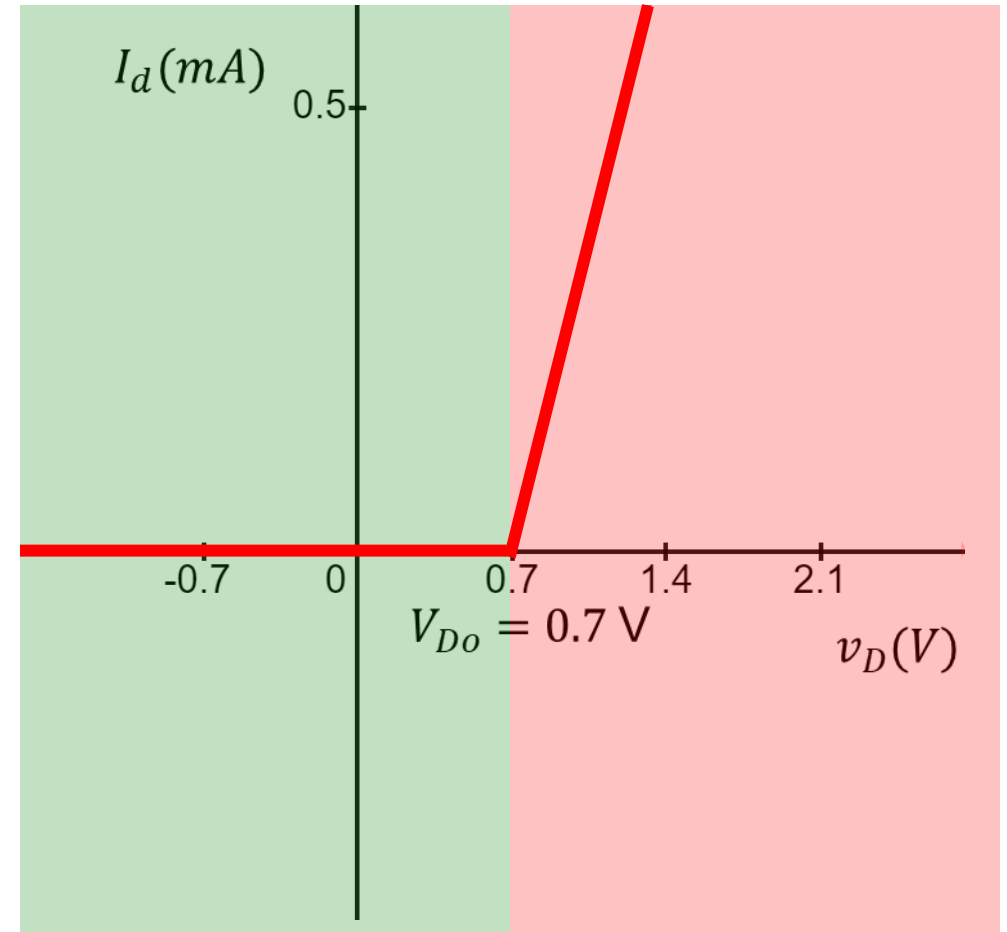
Reverse Bias

Forward Bias



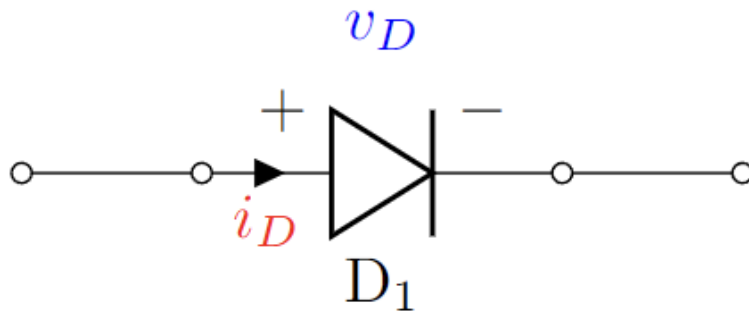
v_D : Total Voltage Across diode

V_{Do} : Diode Cut-off voltage (0.7 V)



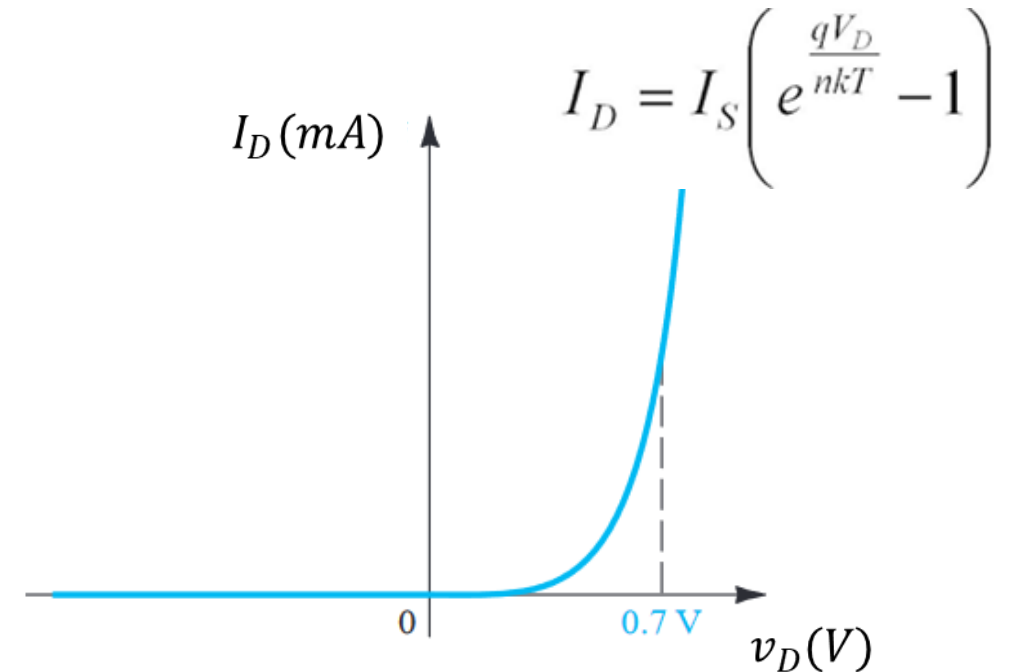
Diode Models

4. Shockley Diode Equation Model:



$$i_D = I_s \left[\exp\left(\frac{v_D}{nV_T}\right) - 1 \right] \quad V_T = \frac{kT}{q}$$

$k = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant and $q = 1.60 \times 10^{-19}$ C is the magnitude of the electrical charge of an electron. At a temperature of 300 K, we have $V_T \cong 26$ mV



Where I_s is reverse saturation current

Solving Circuits with Diodes

- Use **Method of Assumed State!**
- Three steps:
 - **Assume:** One of the **two** operating modes (**Forward Biased – FB** or **Reverse Biased - RB**)

*Correct assumptions will make solving circuit easier.
 - **Solve:** Use corresponding equation and KCL/KVL or any circuit analysis tool of your choice.
 - **Verify:** Check if the conditions of the diodes in the chosen region (**FB or RB**) are satisfied. If not, one can be sure that the other operating mode is correct.

Example Problems (Ideal Diode)

Example 1: $R_L = 2.5 \text{ k}\Omega$

Assume:

Diode is in **FB**: A short circuit,

$$v_d = 0 \text{ V}$$

$$I > 0 \text{ mA}$$

Solve:

$$v_o = v_{IN} = 5 \text{ V}$$

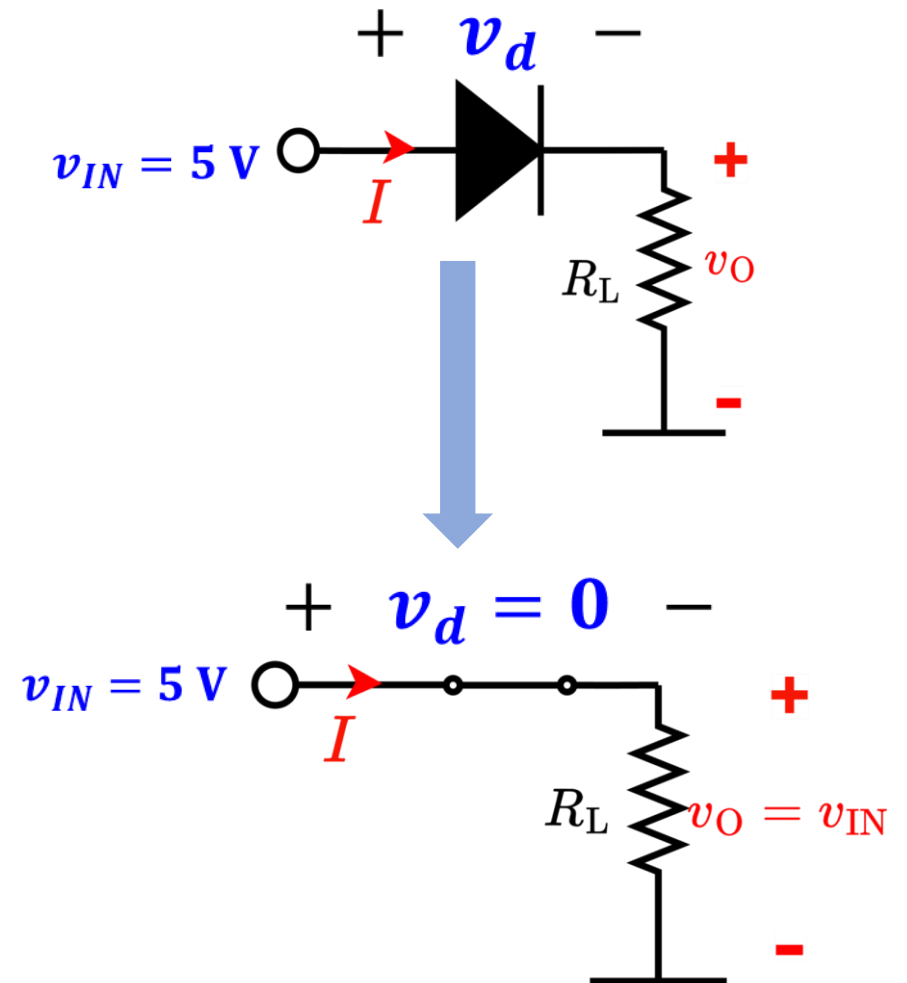
$$v_d = v_{IN} - v_o = 0 \text{ V}$$

$$I = \frac{v_o}{R_L} = \frac{5 \text{ V}}{2.5 \text{ k}\Omega} = 2 \text{ mA} > 0$$

Verify: For **FB** $\rightarrow v_d = 0 \text{ V}$

$$I > 0 \text{ mA}$$

Here, both conditions are fulfilled. **Assumption is Correct!**



Example Problems (Ideal Diode)

Example 2: $R_L = 2.5 \text{ k}\Omega$

Assume:

Diode is in **FB**: A short circuit,

$$v_d = 0 \text{ V}$$

$$I > 0 \text{ mA}$$

Solve:

$$v_o = v_{IN} = 5 \text{ V}$$

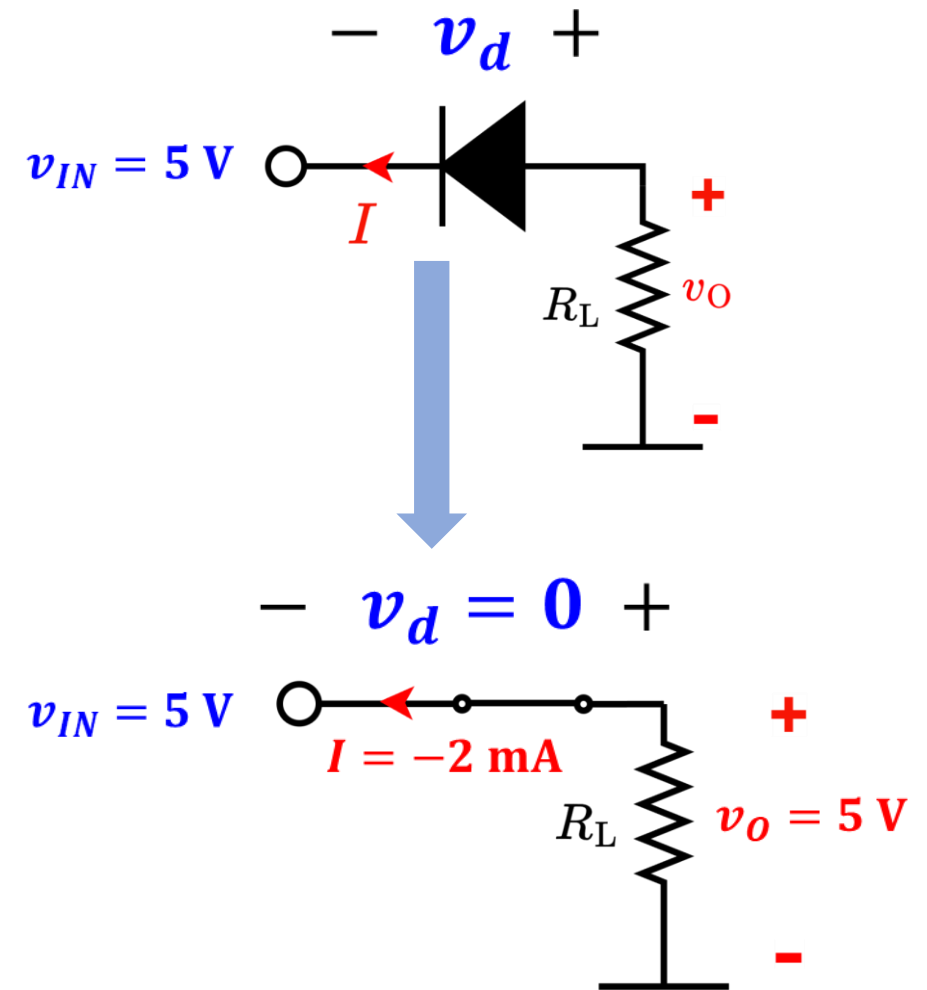
$$v_d = v_o - v_{IN} = 0 \text{ V}$$

$$I = -\frac{v_o}{R_L} = -\frac{5 \text{ V}}{2.5 \text{ k}\Omega} = \boxed{-2 \text{ mA} < 0}$$

Verify: For **FB** $\rightarrow$ $v_d = 0 \text{ V}$

$$I > 0 \text{ mA}$$

Here, $I < 0 \text{ mA}$. **Assumption is INCORRECT!**



Example Problems (Ideal Diode)

Example 2: $R_L = 2.5 \text{ k}\Omega$

Assume:

Diode is in **RB**: An open circuit,

$$v_d < 0 \text{ V}$$

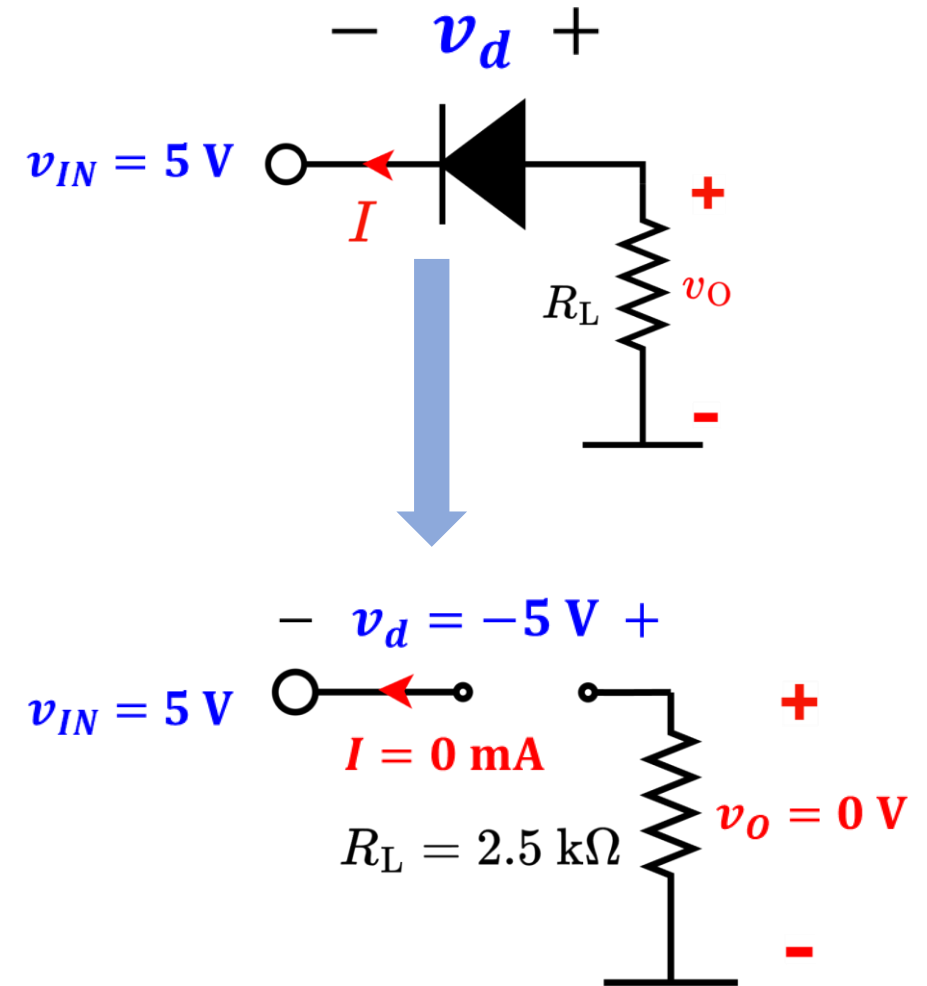
$$I = 0 \text{ mA}$$

Solve:

$$v_o = 0 \text{ V}$$

$$v_d = v_o - v_{IN} = -5 \text{ V} < 0 \text{ V}$$

$$I = 0 \text{ mA}$$



Example Problems (CVD Diode)

Example 3: $R_L = 2.5 \text{ k}\Omega$

Assume:

Diode is in **FB**: A constant voltage source of V_{DO} voltage,

$$v_d = V_{DO} = 0.7 \text{ V}$$

$$I > 0 \text{ mA}$$

Solve:

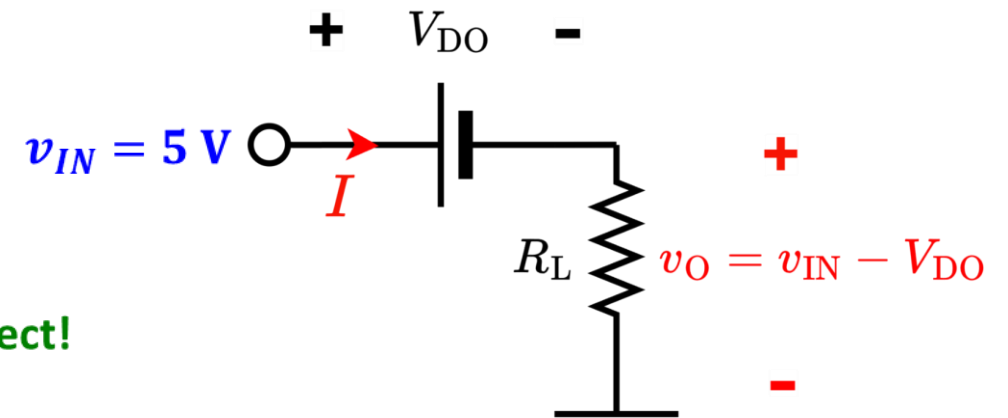
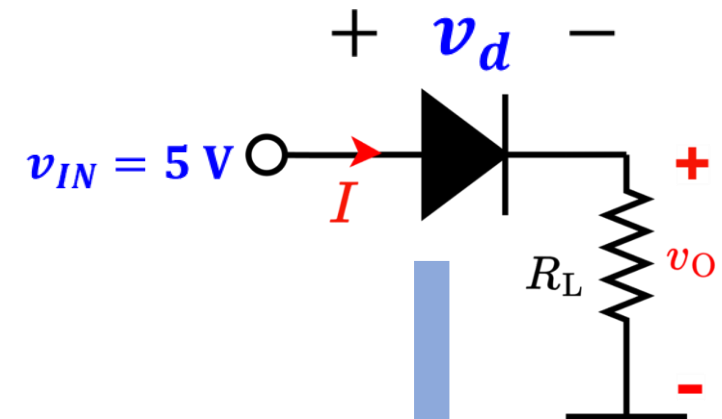
$$v_o = v_{IN} - V_{DO} = 5 - 0.7 \text{ V} = 4.3 \text{ V}$$

$$v_d = V_{DO} = 0.7 \text{ V}$$

$$I = \frac{v_{IN} - V_{DO}}{R_L} = \frac{4.3 \text{ V}}{2.5 \text{ k}\Omega} = 1.72 \text{ mA} > 0$$

Verify: For **FB** $\rightarrow I > 0 \text{ mA}$

Here, both conditions are fulfilled. **Assumption is Correct!**



Example Problems (CVD Diode)

Example 4: $R_L = 2.5 \text{ k}\Omega$

Assume:

Diode is in **FB**: A constant voltage source of V_{DO} voltage,

$$v_d = V_{DO} = 0.7 \text{ V}$$

$$I > 0 \text{ mA}$$

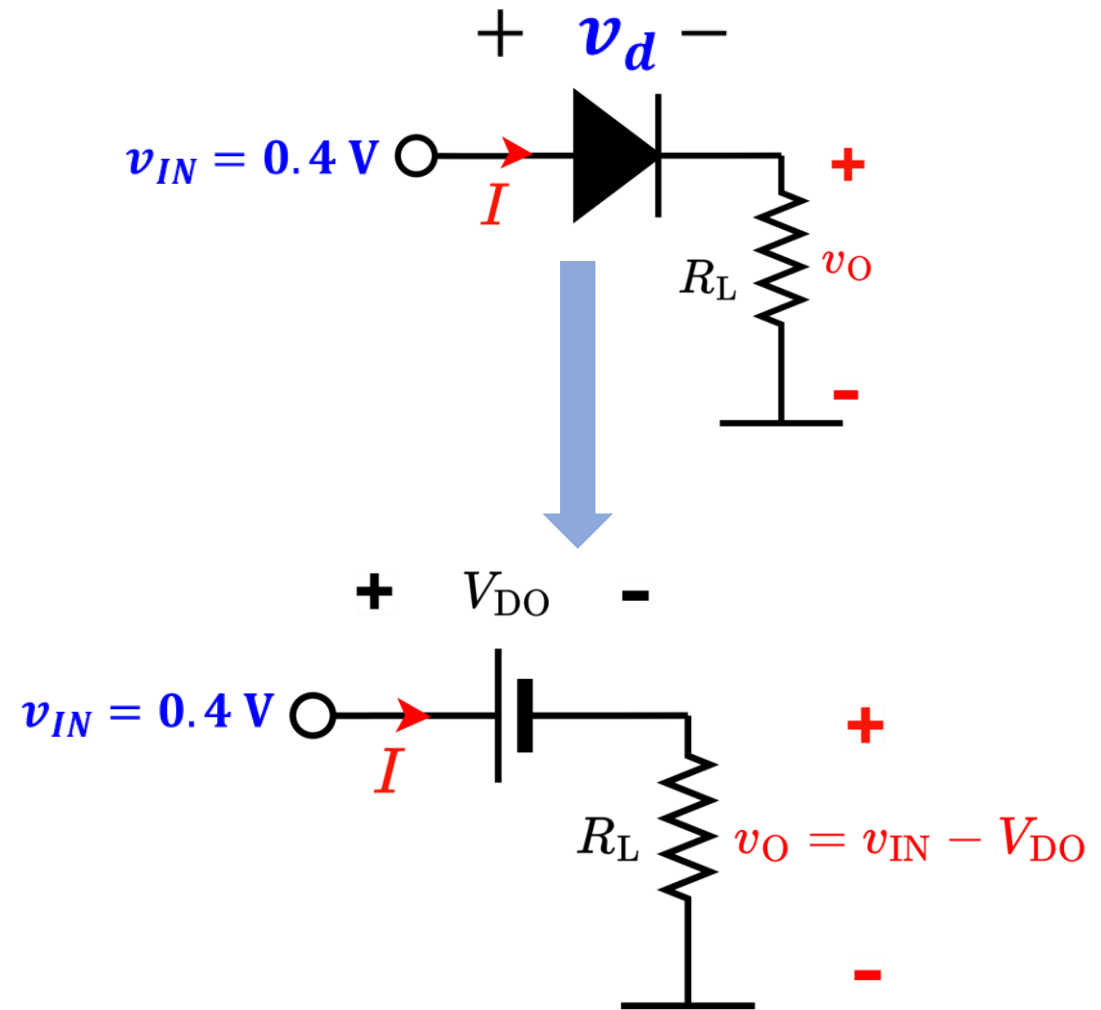
Solve:

$$v_o = v_{IN} - V_{DO} = 0.4 - 0.7 \text{ V} = -0.3 \text{ V}$$

$$I = \frac{v_o}{R_L} = \frac{-0.3 \text{ V}}{2.5 \text{ k}\Omega} = \boxed{-0.12 \text{ mA} < 0}$$

Verify: For **FB** $\rightarrow I > 0 \text{ mA}$

Here, $I < 0 \text{ mA}$. **Assumption is INCORRECT!**



Example Problems (Ideal Diode)

Example 4: $R_L = 2.5 \text{ k}\Omega$

Assume:

Diode is in **RB**: An open circuit,

$$v_d < V_{DO} = 0.7 \text{ V}$$

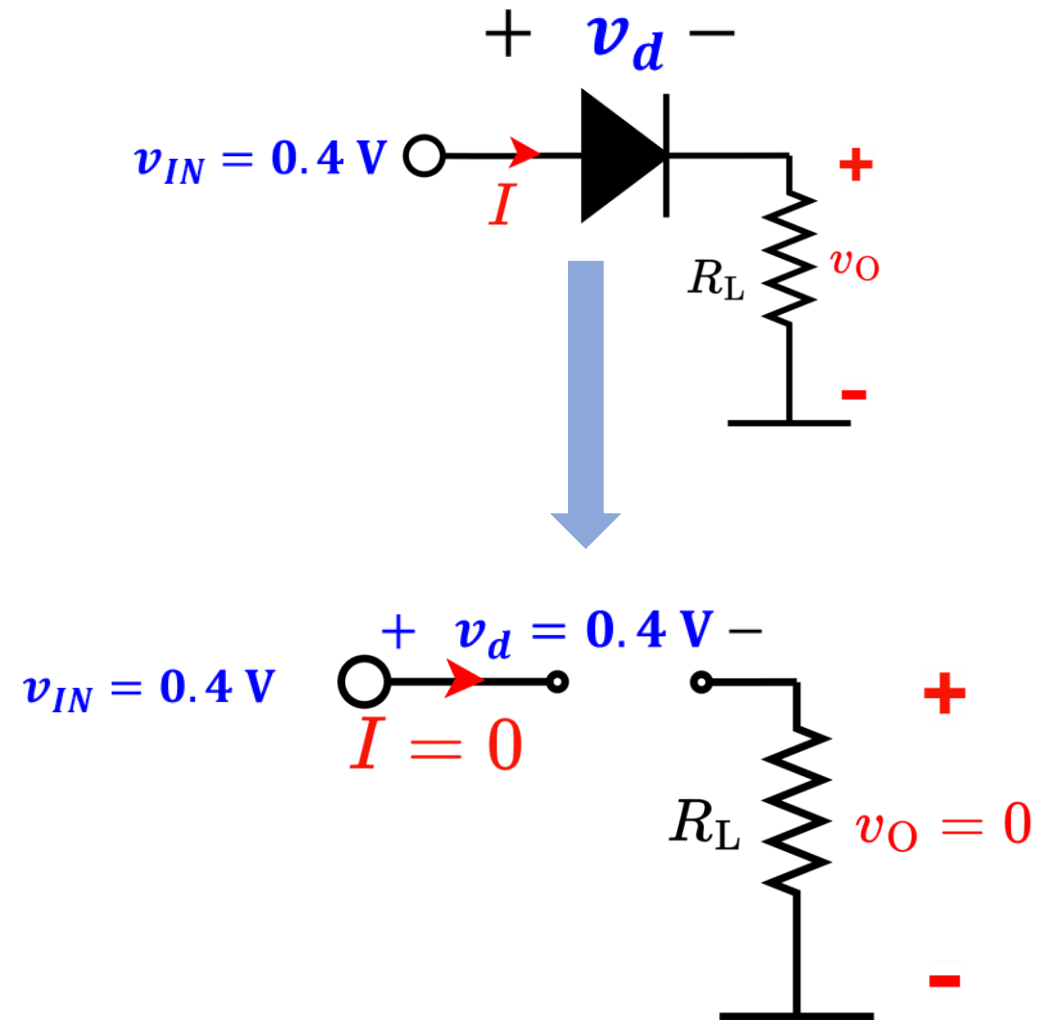
$$I = 0 \text{ mA}$$

Solve:

$$I = 0 \text{ mA}$$

$$v_o = 0 \text{ V}$$

$$v_d = v_{IN} - v_o = 0.4 \text{ V} < 0.7 \text{ V}$$



Example Problems (CVD+R Diode)

Example 5: $R_L = 2.5 \text{ k}\Omega$, $V_{DO} = 0.7 \text{ V}$, $r_d = 50 \Omega$

Assume:

Diode is in **FB**: A constant voltage drop in series with a resistor,

$$V_{DO} = 0.7 \text{ V}, \quad r_d = 50 \Omega$$

$$v_d = V_{DO} + I r_d$$

$$I > 0 \text{ mA}$$

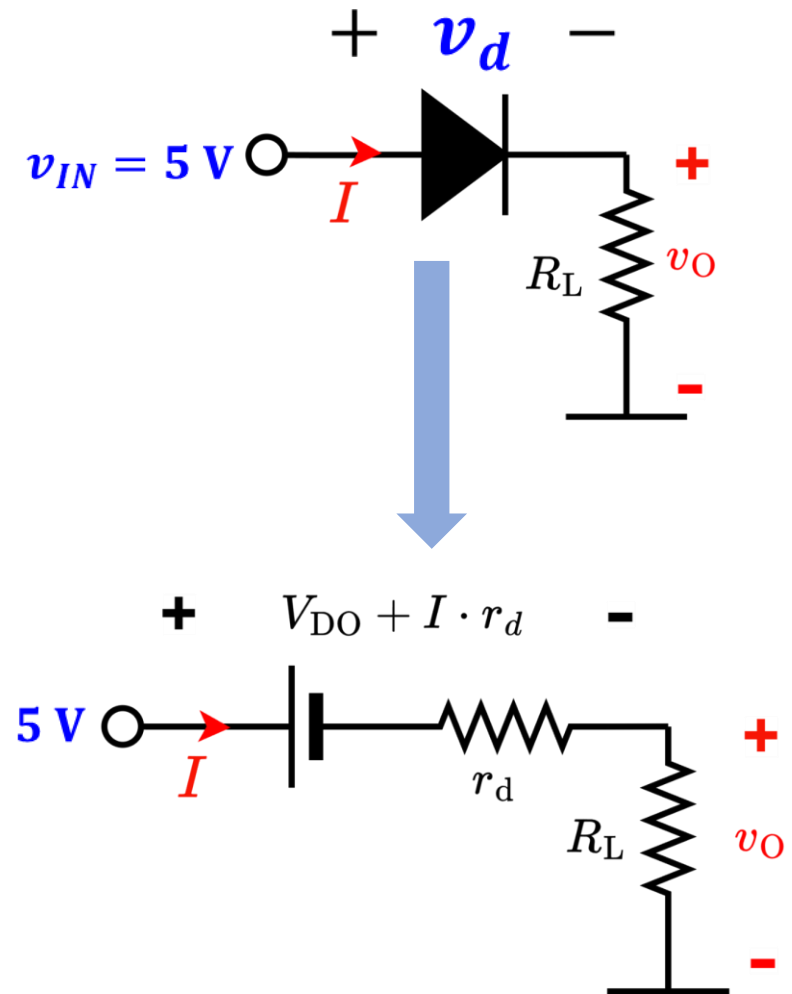
Solve:

$$I = \frac{v_O - V_{DO}}{R_L + r_d} = \frac{4.3 \text{ V}}{2.5 \text{ k}\Omega + 0.05 \text{ k}\Omega} = 1.70 \text{ mA}$$

$$v_O = I R_L = 1.70 \times 2.5 \text{ V} = 4.25 \text{ V}$$

Verify: For **FB** $\rightarrow I > 0 \text{ mA}$

Here, the condition is fulfilled. **Assumption is Correct!**



Example Problems (CVD+R Diode)

Example 6: $R_L = 2.5 \text{ k}\Omega$, $V_{DO} = 0.7 \text{ V}$, $r_d = 50 \Omega$

Assume:

Diode is in **RB**: An open circuit,

$$v_d < V_{DO}$$

$$I = 0 \text{ mA}$$

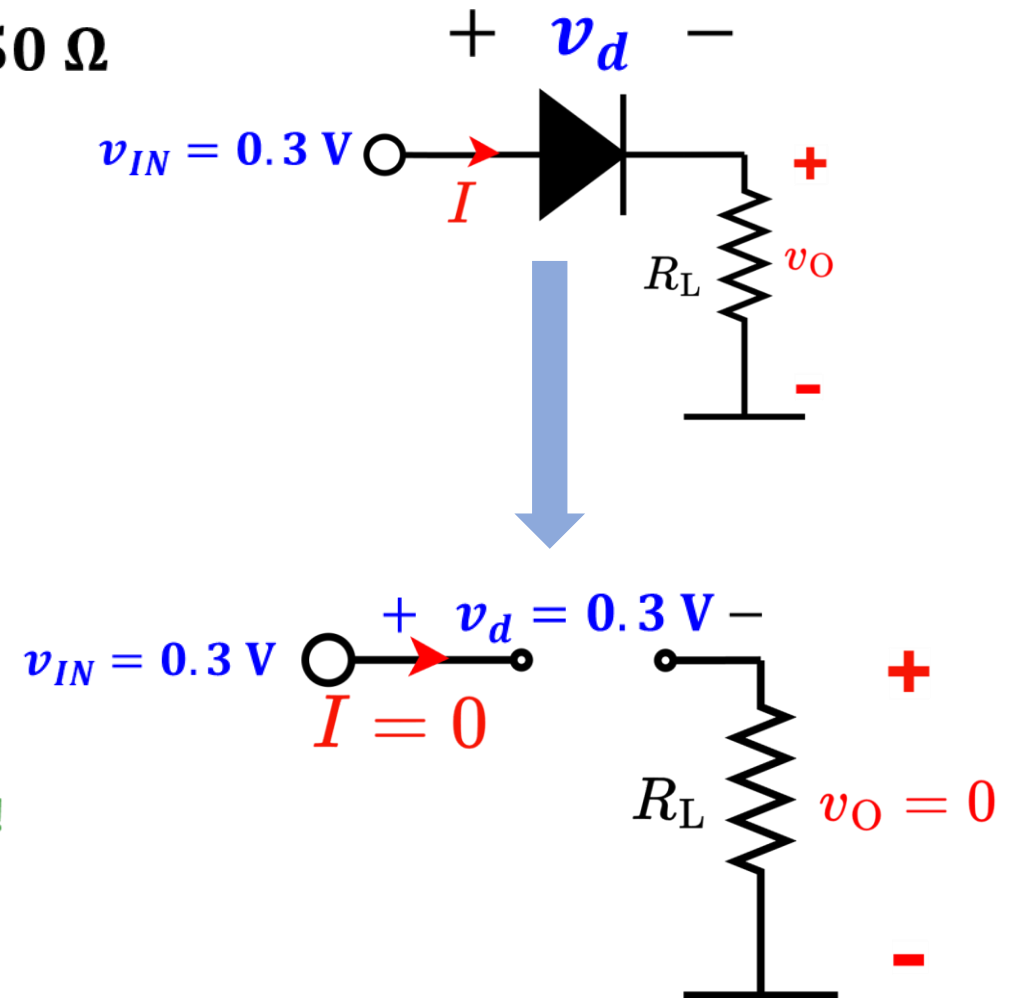
Solve:

$$v_o = 0 \text{ V}$$

$$v_d = v_{IN} - v_o = 0.3 \text{ V}$$

Verify: For **FB** $\rightarrow v_d < V_{DO}$

Here, the condition is fulfilled. **Assumption is Correct!**

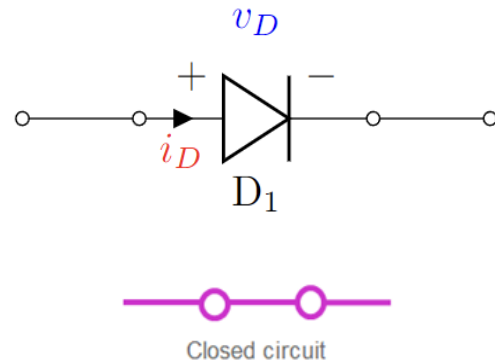


Diode Models: Summary

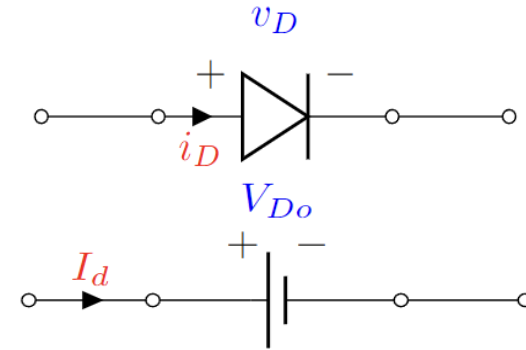
Ideal Diode

$$i_D > 0$$

Forward Bias



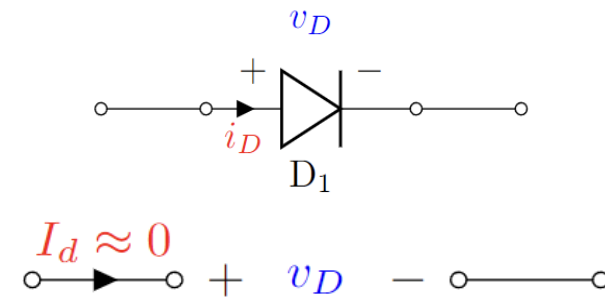
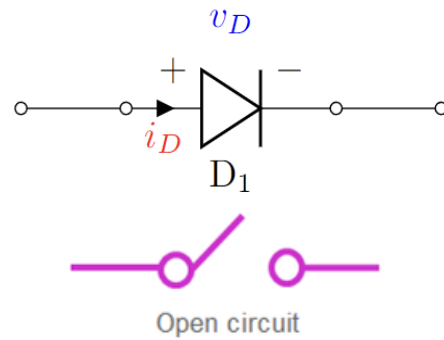
CVD Diode



$$v_D < V_{D0}$$

$$i_D = 0$$

Reverse Bias

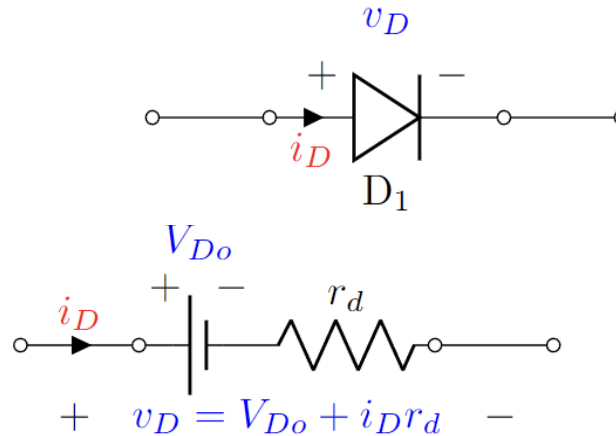


Diode Models: Summary

CVD+R Diode

$$i_D > 0$$

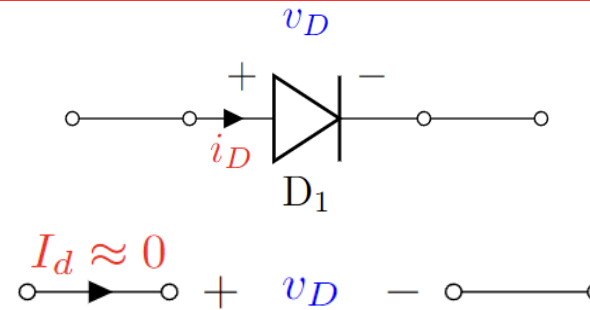
Forward Bias



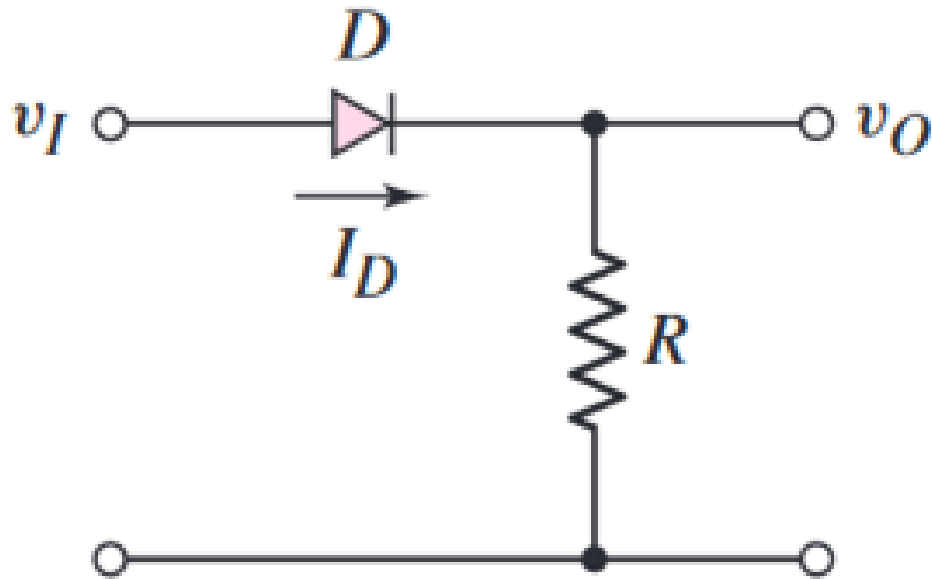
$$v_D < V_{D0}$$

$$i_D = 0$$

Reverse Bias

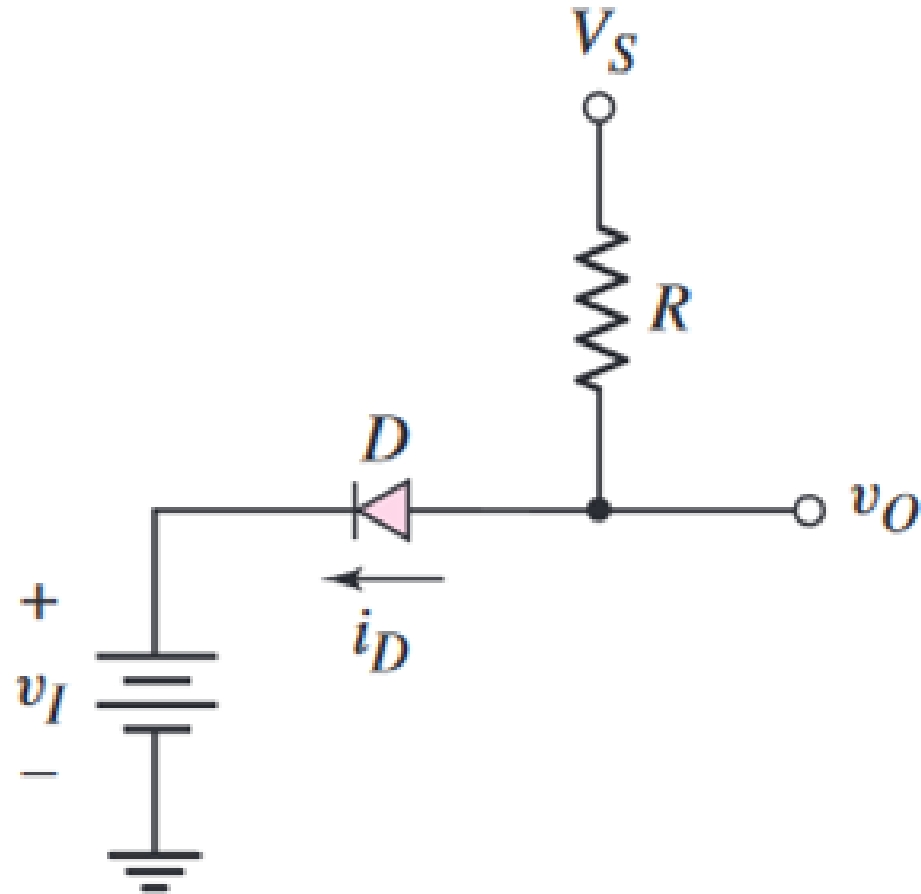


Solving Circuits with diodes (7)



If $v_I = 5 \text{ V}$ and $V_{DO} = 0.7 \text{ V}$, find v_O

Solving Circuits with diodes (7)

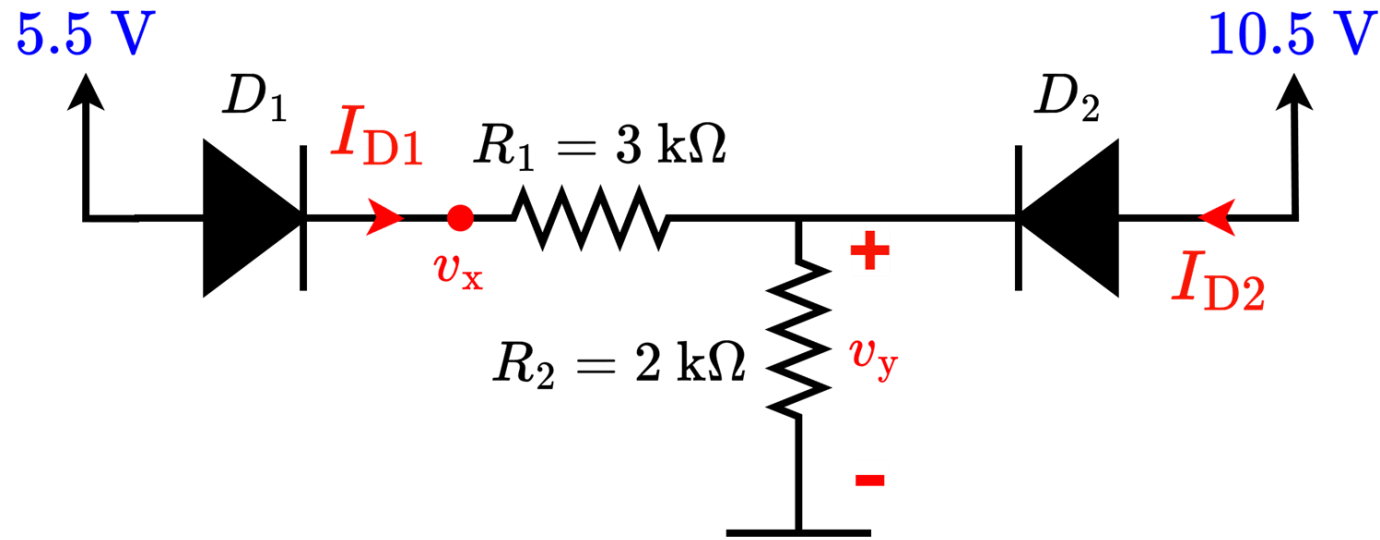


i) If $v_I = 5 \text{ V}$, $V_S = 10 \text{ V}$ and $V_{DO} = 0.7 \text{ V}$, find v_O

ii) If $v_I = 11 \text{ V}$, $V_S = 10 \text{ V}$ and $V_{DO} = 0.7 \text{ V}$, find v_O

iii) If $v_I = 0 \text{ V}$, $V_S = 10 \text{ V}$ and $V_{DO} = 0.7 \text{ V}$, find v_O

Solving Circuits with diodes (9)



Example 9: Analyze the circuit to find I_{D1} , I_{D2} , v_x and v_y . Consider $V_{DO} = 0.5 \text{ V}$.
[Validate Assumptions]

Solving Circuits with diodes (9)

Example 9: Analyze the circuit to find I_{D1} , I_{D2} , v_x and v_y . Consider $V_{D0} = 0.5 \text{ V}$.

[Validate Assumptions]

Assume:

D1, D2 are in FB: A constant voltage source of V_{D0} voltage,

$$\begin{array}{ll} v_{D1} = 0.5 \text{ V}, & || \\ I_{D1} > 0 \text{ mA}, & || \\ v_{D2} = 0.5 \text{ V} & \\ I_{D2} > 0 \text{ mA} & \end{array}$$

Solve:

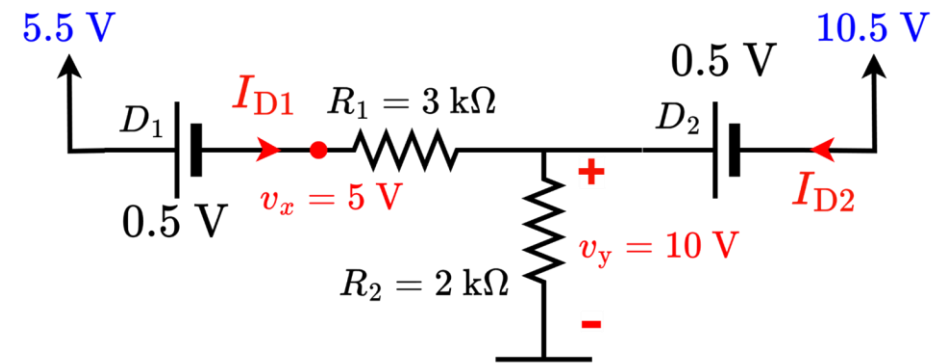
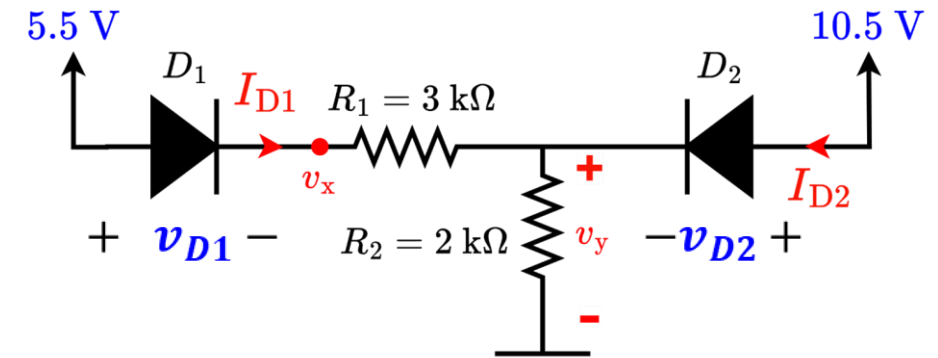
$$v_x = 5.5 - v_{D1} = 5 \text{ V}$$

$$v_y = 10.5 - v_{D2} = 10 \text{ V}$$

$$I_{D1} = \frac{v_x - v_y}{R_1} = \frac{-5 \text{ V}}{3 \text{ k}\Omega} = -1.66 \text{ mA} < 0$$

$$I_{D2} = \frac{v_y}{R_2} - I_{D1} = \frac{10}{2} - \frac{-5}{3} = \frac{20}{3} \text{ mA} > 0$$

Verify: $I_{D1} > 0 \text{ mA}$, $I_{D2} > 0 \text{ mA}$
Here, $I_{D1} < 0 \text{ mA}$. **Assumption is INCORRECT!**



Solving Circuits with diodes (9)

Example 9: Analyze the circuit to find I_{D1} , I_{D2} , v_x and v_y . Consider $V_{D0} = 0.5 \text{ V}$.

[Validate Assumptions]

Assume:

D1 in RB

$$I_{D1} = 0 \text{ mA},$$

$$v_{D1} < 0.5 \text{ V},$$

D2 are in FB

$$v_{D2} = 0.5 \text{ V}$$

$$I_{D2} > 0 \text{ mA}$$

Solve:

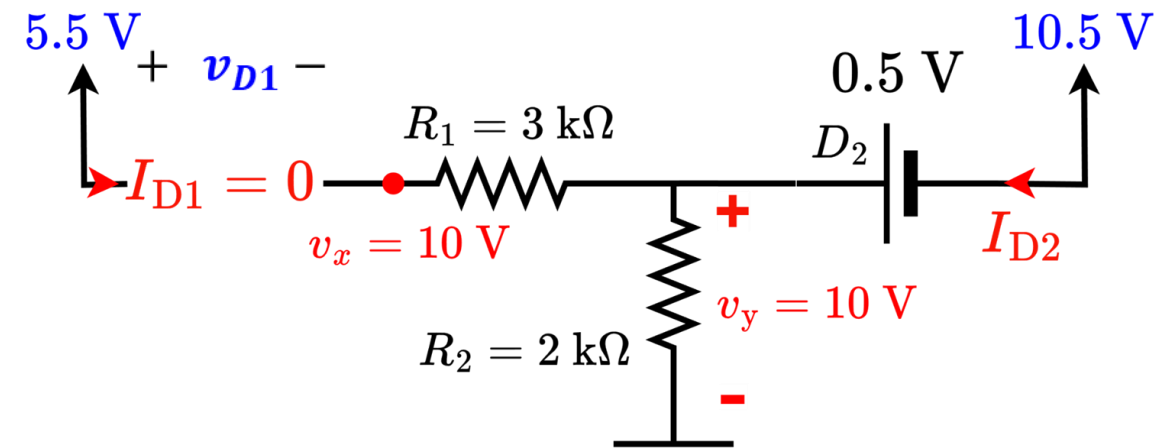
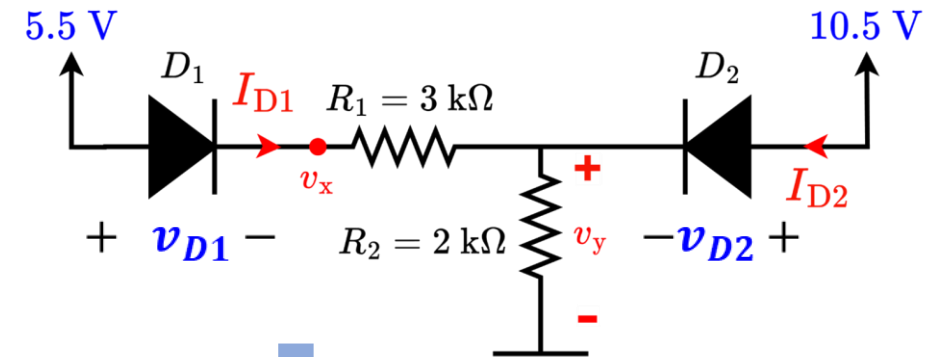
$$v_y = 10.5 - v_{D2} = 10 \text{ V}$$

$$v_x = v_y = 10 \text{ V, as } I_{D1} = 0$$

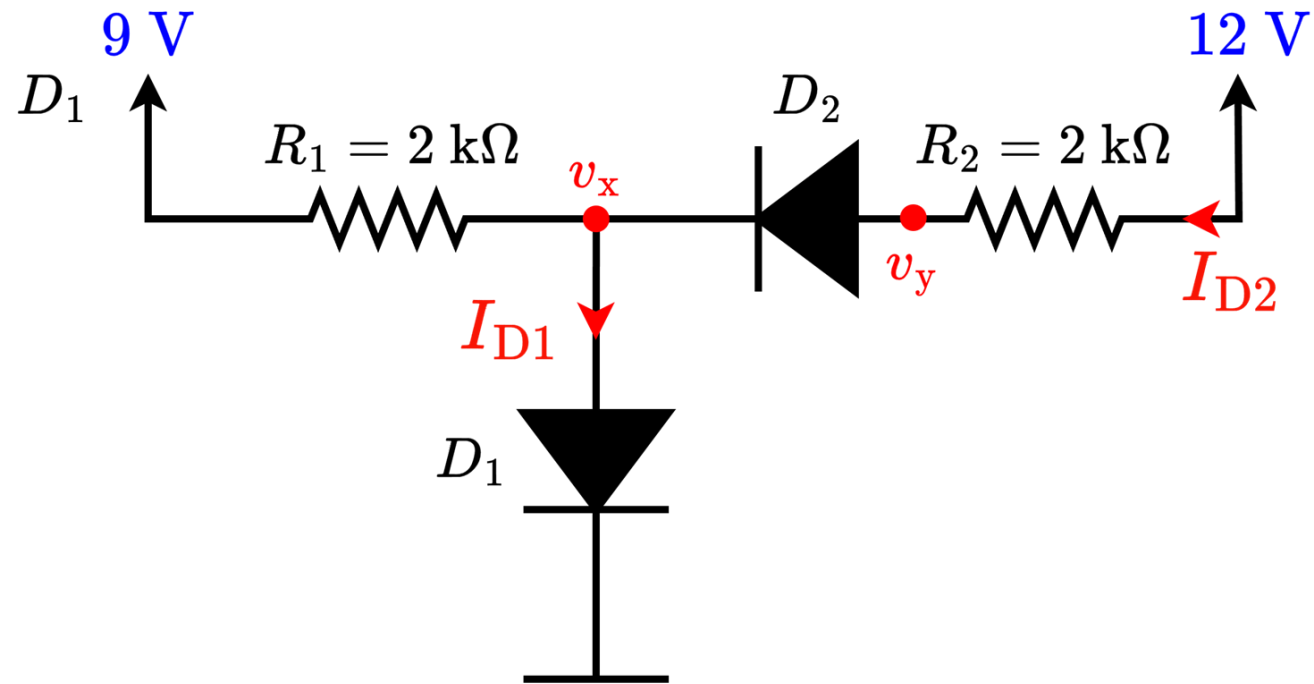
$$v_{D1} = 5.5 - v_x = -4.5 \text{ V} < 0.5 \text{ V}$$

$$I_{D2} = \frac{v_y}{R_2} = \frac{10}{2} = 5 \text{ mA} > 0$$

Verify: $v_{D1} < 0.5 \text{ V}$, $I_{D2} > 0 \text{ mA}$
Assumption is Correct!



Solving Circuits with diodes (10)



Example 10: Analyze the circuit to find I_{D1} , I_{D2} , v_x and v_y . Consider $V_{D0} = 1\text{ V}$. **[Validate Assumptions]**. Are the diodes consuming power or delivering power?

Solving Circuits with diodes (10)

Example 10: Analyze the circuit to find I_{D1} , I_{D2} , v_x and v_y . Consider $V_{D0} = 1\text{ V}$. [Validate Assumptions]. Are the diodes consuming power or delivering power?

Assume:

D1, D2 are in FB: A constant voltage source of V_{D0} voltage,

$$\begin{array}{lcl} v_{D1} = 1\text{ V}, & || & v_{D2} = 1\text{ V} \\ I_{D1} > 0\text{ mA}, & || & I_{D2} > 0\text{ mA} \end{array}$$

Solve:

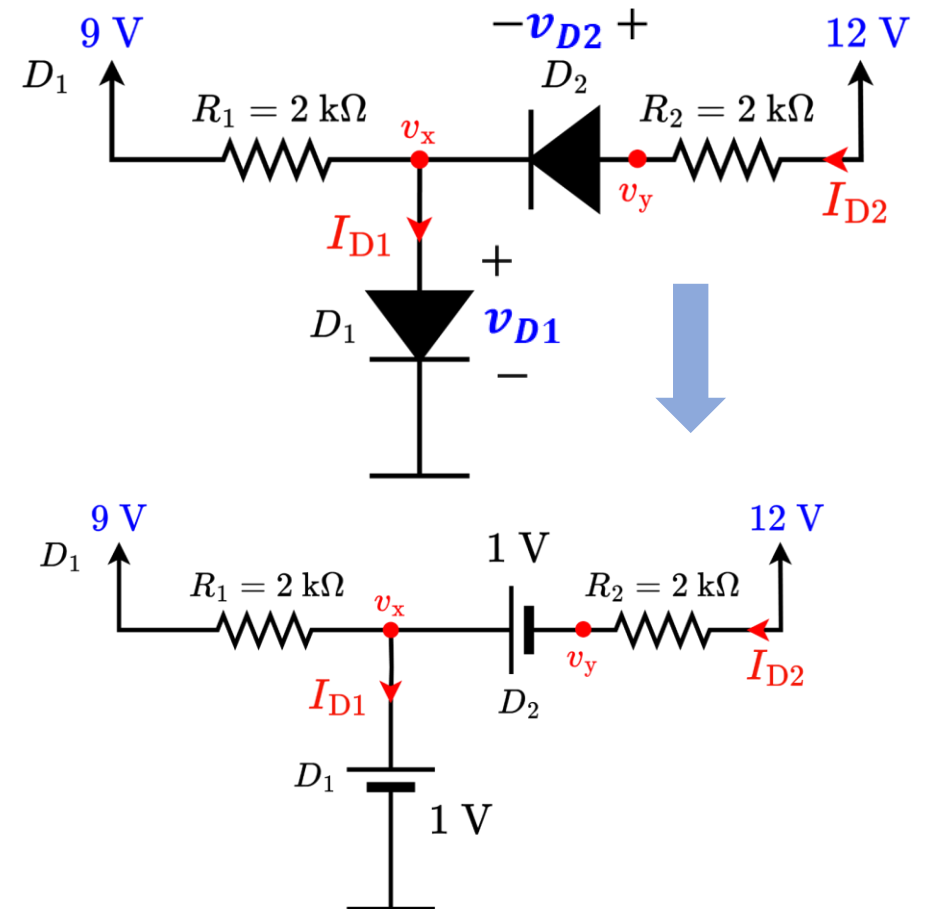
$$v_x = v_{D1} = 1\text{ V}$$

$$v_y = v_x + v_{D2} = 2\text{ V}$$

$$I_{D2} = \frac{12 - v_y}{R_2} = \frac{10}{2} = 5\text{ mA} > 0$$

$$I_{D1} = I_{D2} + \frac{9 - v_x}{R_1} = 5 + \frac{8}{2} = 9\text{ mA} > 0$$

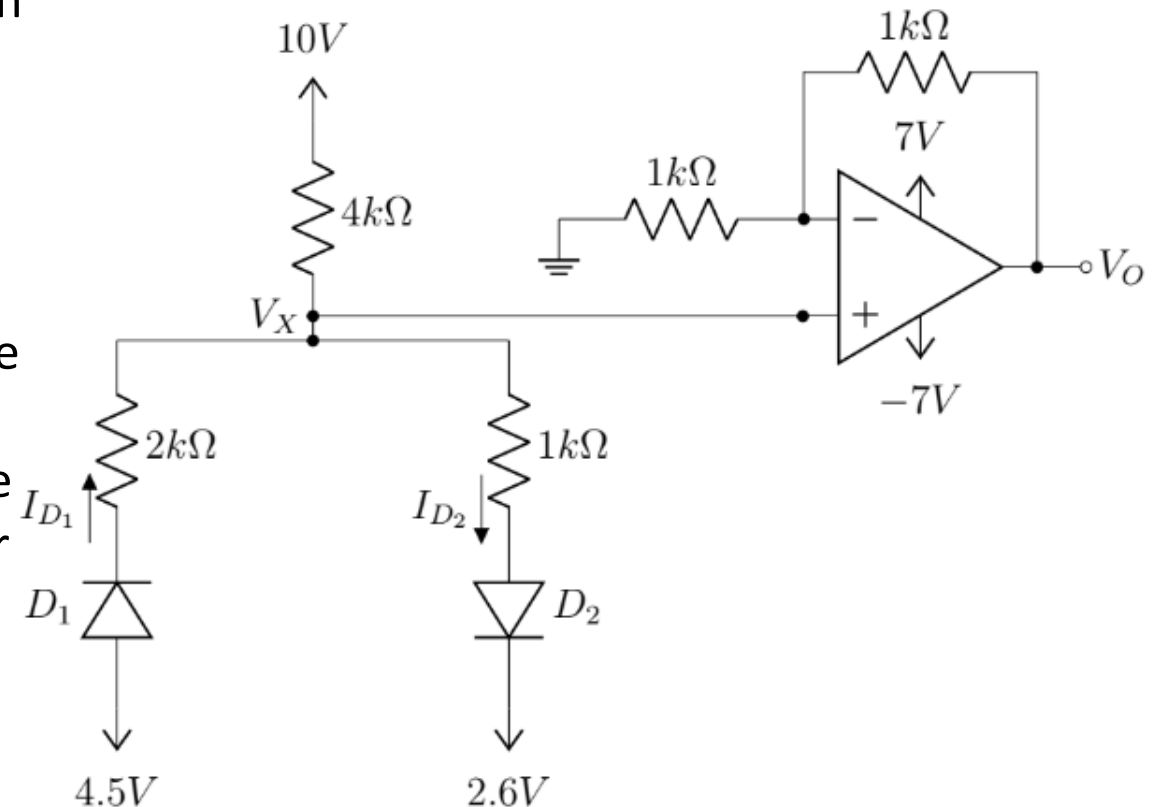
Verify: $I_{D1} > 0\text{ mA}$, $I_{D2} > 0\text{ mA}$
Assumption is Correct!



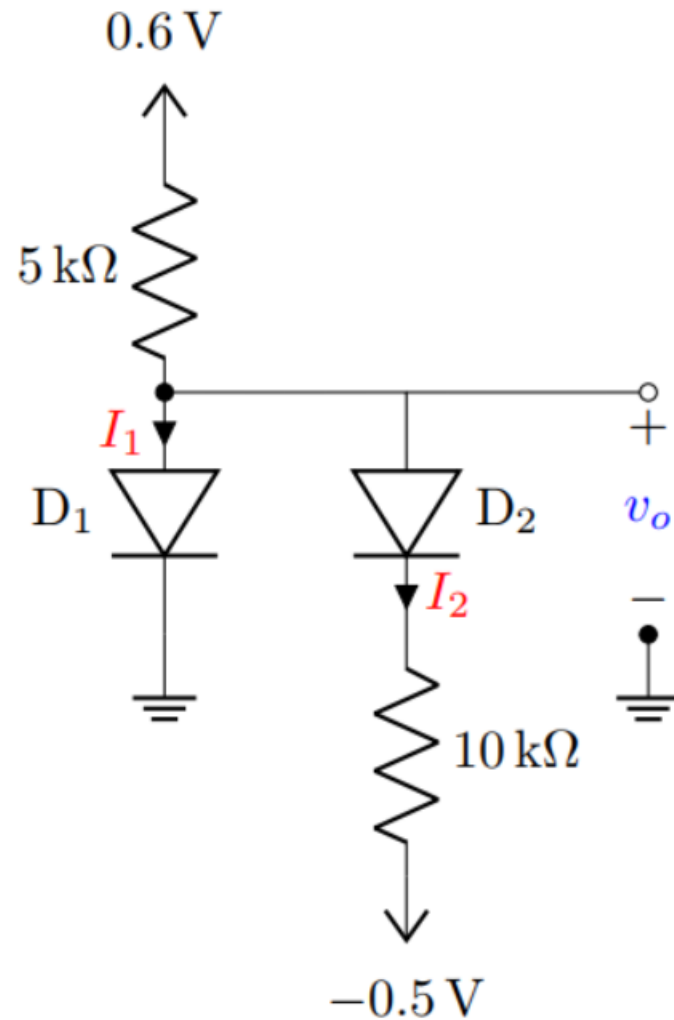
Solving Circuits with diodes (11)

Consider the following circuit with an ideal OP-Amp and two diodes. The Diode D_1 and D_2 have V_{D0} of 0.5V and 0.6V respectively.

- a) State the equivalent circuit model of an ideal diode in reverse bias.
- b) Analyze the circuit and calculate V_x , I_{D1} and I_{D2} . Use method of assumed states. You must validate your assumption.
- c) Determine the output voltage V_o .



Solving Circuits with diodes (12)

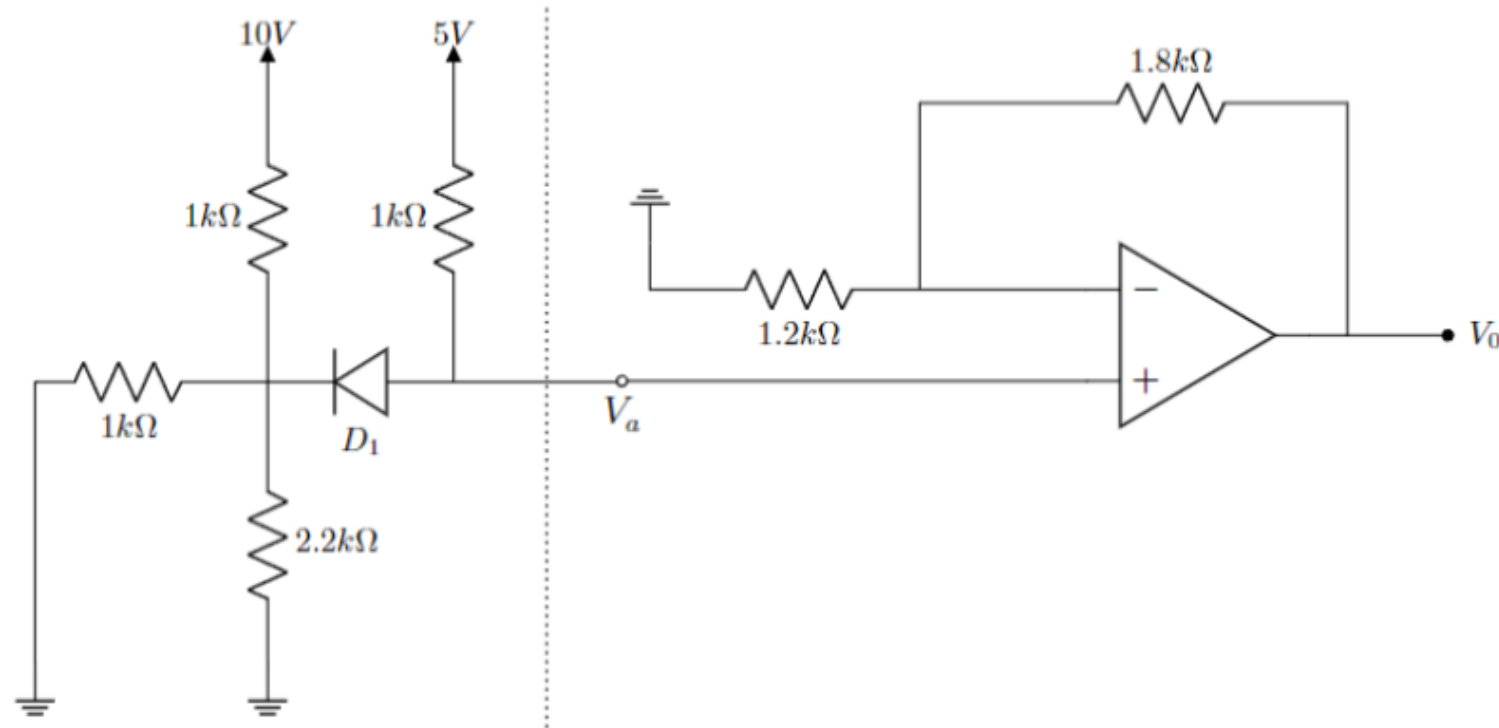


Analyze the circuit to find I_1 , I_2 , and v_o for

- Assuming all diodes as ideal
- Consider $V_{D1} = 0.5\text{ V}$ and $V_{D2} = 0.7\text{ V}$. (CVD model)
- Consider $V_{D0} = 0.6\text{ V}$ and $r_d = 50\text{ }\Omega$. (CVD+R model)

[Validate Assumptions]

Solving Circuits with diodes (13)

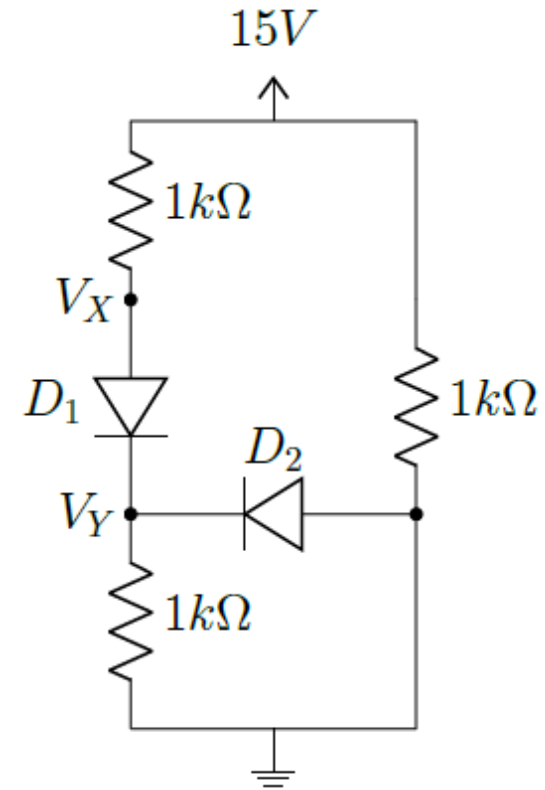


The saturation voltages of the Op-Amp are given as- $V_{sat}^+ = +10V$ and $V_{sat}^- = -10V$. The forward voltage drop of the diode, V_D is $0.7V$.

- (a) **Determine** the operating mode diode, D_1 . Verify your assumption with necessary calculations.
- (b) **Calculate** the voltage at - (i) node 'Va', (ii) non-inverting terminal of the Op-Amp, (iii) inverting terminal of the Op-Amp.
- (c) Find out the output voltage, V_o of the Op-Amp.

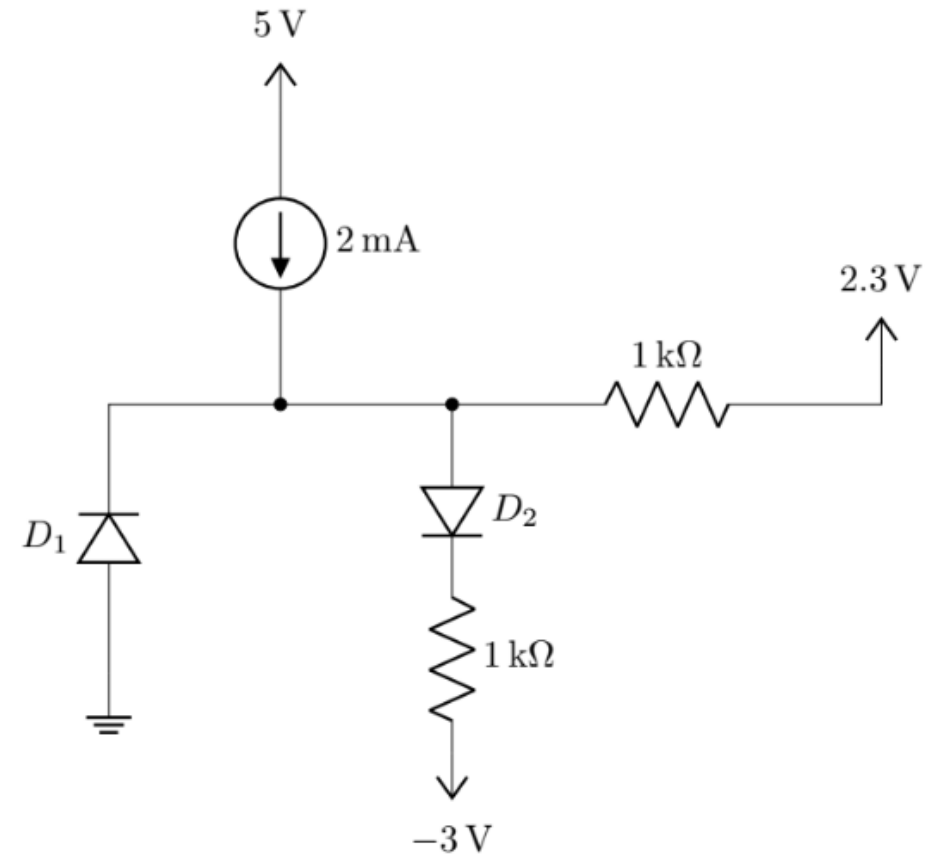
Solving Circuits with diodes (14)

Analyze the circuit to calculate V_X , V_Y , I_{D1} and I_{D2} using the method of assumed states. You must validate your assumptions. Use $V_{DO} = 0.7V$ for both diodes.



Solving Circuits with diodes (15)

Analyze the circuit to determine which diode(s) are conducting. Additionally, calculate the power dissipated by any conducting diode(s). Use $V_{D0} = 0.7\text{V}$ for both diodes.



Thank You!